

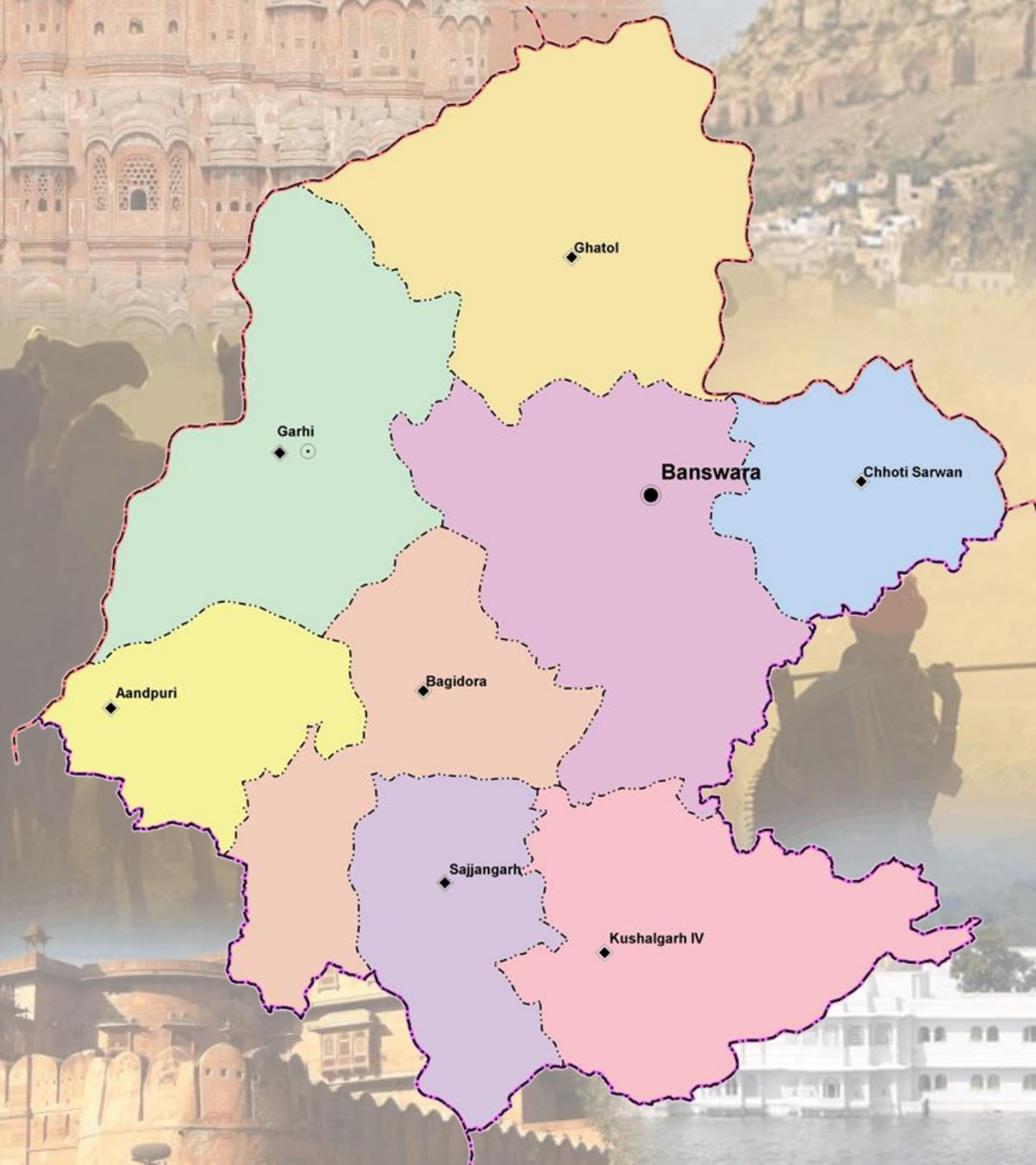


Ground Water Department,
Rajasthan

Hydrogeological Atlas of Rajasthan Banswara District



European Union
State Partnership Programme



2013



Rolta India Limited

Message



Shri Ashok Gehlot,
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

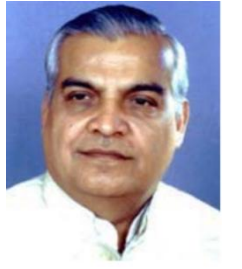
It is also heartening to note that the outcome of the project is being published in the form of 'Hydrogeological Atlas of Rajasthan'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.



(Ashok Gehlot)

Message

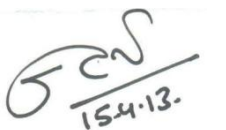


Dr. Jitendra Singh,
Hon'ble Minister for Energy,
Non-Conventional Energy Resources,
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.



(Jitendra Singh)

Message



Dr. Purushottam Agarwal

Principal Secretary to Government,

Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of 'Hydrogeological Atlas of Rajasthan' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

(Purushottam Agarwal)

Foreword



Arno Schaefer

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

(Arno Schaefer)

Hydrogeological Atlas of Rajasthan

Banswara District



Contents:

List of Plates	Title	Page No.
Plate I	Administrative Map	2
Plate II	Topography	4
Plate III	Rainfall Distribution	4
Plate IV	Geological Map	6
Plate V	Geomorphological Map	6
Plate VI	Aquifer Map	8
Plate VII	Stage of Ground Water Development (Block wise) 2011	8
Plate VIII	Location of Exploratory and Ground Water Monitoring Stations	10
Plate IX	Depth to Water Level (Pre-Monsoon 2010)	10
Plate X	Water Table Elevation (Pre-Monsoon 2010)	12
Plate XI	Water Level Fluctuation (Pre-Post Monsoon 2010)	12
Plate XII	Electrical Conductivity Distribution (Average Pre-Monsoon 2005-09)	14
Plate XIII	Chloride Distribution (Average Pre-Monsoon 2005-09)	14
Plate XIV	Fluoride Distribution (Average Pre-Monsoon 2005-09)	16
Plate XV	Nitrate Distribution (Average Pre-Monsoon 2005-09)	16
Plate XVI	Depth to Bedrock	18
Plate XVII	Map of Unconfined Aquifer	18
Glossary of terms		19

ADMINISTRATIVE SETUP

DISTRICT – BANSWARA

Location:

Banswara district is located in the extreme southern part of Rajasthan. It is bounded in the North by Dungarpur and Pratapgarh districts, in the west by Dungarpur district, in the East by state of Madhya Pradesh and South by state of Gujarat and. It stretches between 23°03' 22.98" to 23° 55' 31.36" north latitude and 73° 57' 12.92" to 74° 46' 22.15" east longitude covering area of 4,493 sq km. Major part of the district has a systematic drainage system, as whole region is the part of 'Mahi River Basin'.

Administrative Set-up:

Banswara district is administratively divided into ten Blocks. The following table summarizes the basic statistics of the district at block level.

S. No.	Block Name	Population (Based on 2001 census)	Area (sq km)	% of District Area	Total Number of Towns and Villages
1	Anandpuri	1,11,206	347.8	7.7	129
2	Bagidora	1,76,729	512.2	11.4	165
3	Banswara	2,99,392	771.5	17.2	224
4	Chhoti Sarwan	70,285	374.2	8.3	103
5	Garhi	2,47,468	676.8	15.1	171
6	Ghatol	2,30,344	797.7	17.8	223
7	Kushalgarh	1,49,078	633.4	14.1	211
8	Sajjangarh	1,34,456	379.4	8.4	187
Total		14,18,958	4,493.0	100.0	1,413

Banswara district has 1,413 towns and villages, out of which eight are block headquarters as well.

Climate:

The climate of the district is dry except during the period of SW monsoon. The winter season is from November to February and is followed by summer from March to June. Maximum temperature is between 45 °C to 46 °C in summer and minimum temperature is between 8°C to 12°C in winter. From mid of September to end of November constitute post monsoon season. Average annual rainfall of the district is 893.8mm.

TOPOGRAPHY

Topographically Banswara district represents a rugged terrain. The eastern part of district is occupied by flat-topped hills of the Deccan trap. The drainage system belongs to the Mahi river basin. Its main tributaries are Anas, Chap, Haran, Kagdi & Nal. There are scattered ranges of Aravalis in the eastern half of the district. Minimum elevation (113 m) is found in Anandpuri block whereas highest elevation is reached (581 m) in Kushalgarh block.

Table: Block wise minimum and maximum elevation

S. No.	Block Name	Minimum Elevation (m amsl)	Maximum Elevation (m amsl)
1	Anandpuri	113.0	323.9
2	Bagidora	122.0	456.5
3	Banswara	159.9	508.5
4	Chhoti Sarwan	227.5	564.9
5	Garhi	113.0	292.9
6	Ghatol	140.6	480.8
7	Kushalgarh	218.6	581.0
8	Sajjangarh	158.0	405.0

RAINFALL

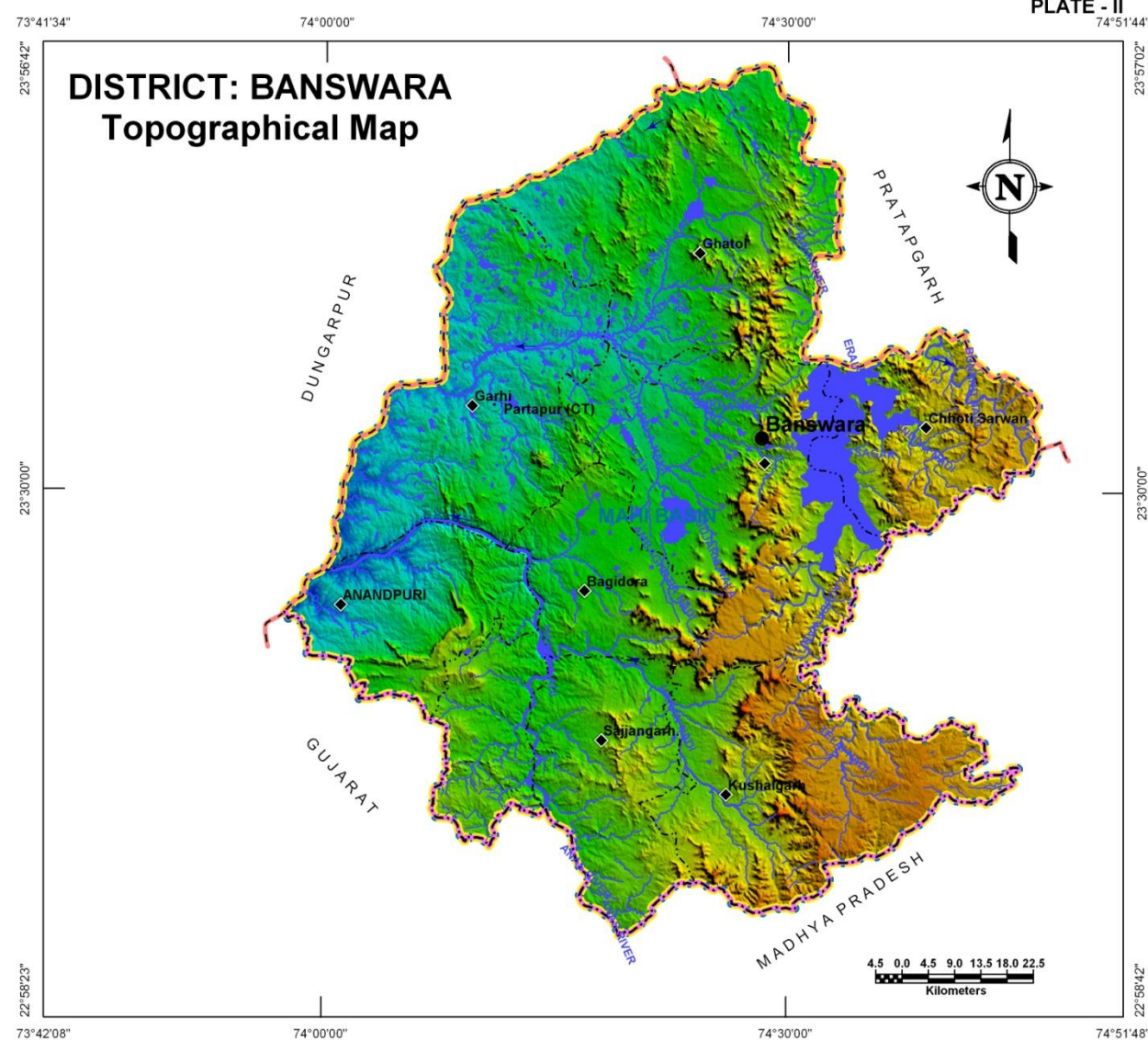
Rainfall received by the district is fairly good. The general distribution of rainfall across can be visualized from isohyets presented in Plate III. Here, southeastern part receives more rainfall as compares to rest of the district and its gradually decreasing towards northwest. In northwestern part (like Garhi) received maximum annual rainfall (524 mm) whereas Kushalgarh receive more Monsoon rainfall (1,011 mm) in southern part of the district. Average annual rainfall in Banswara district was about 690.7 mm as per the data from available rain gauge stations in the year 2010. The highest average annual rainfall noticed in Kushalgarh block.

Table: Block wise annual rainfall statistics (derived from year 2010 meteorological station data)

Block Name	Minimum Annual Rainfall (mm)	Maximum Annual Rainfall (mm)	Average Annual Rainfall (mm)
Anandpuri	584.1	645.3	613.9
Bagidora	596.5	785.9	665.2
Banswara	567.4	875.9	680.3
Chhoti Sarwan	613.4	736.1	665.4
Garhi	524.8	616.9	574.6
Ghatol	584.6	667.8	618.6
Kushalgarh	779.7	1,011.9	928.1
Sajjangarh	659.0	926.0	779.4



PLATE - II



LEGEND

Admin Boundary:

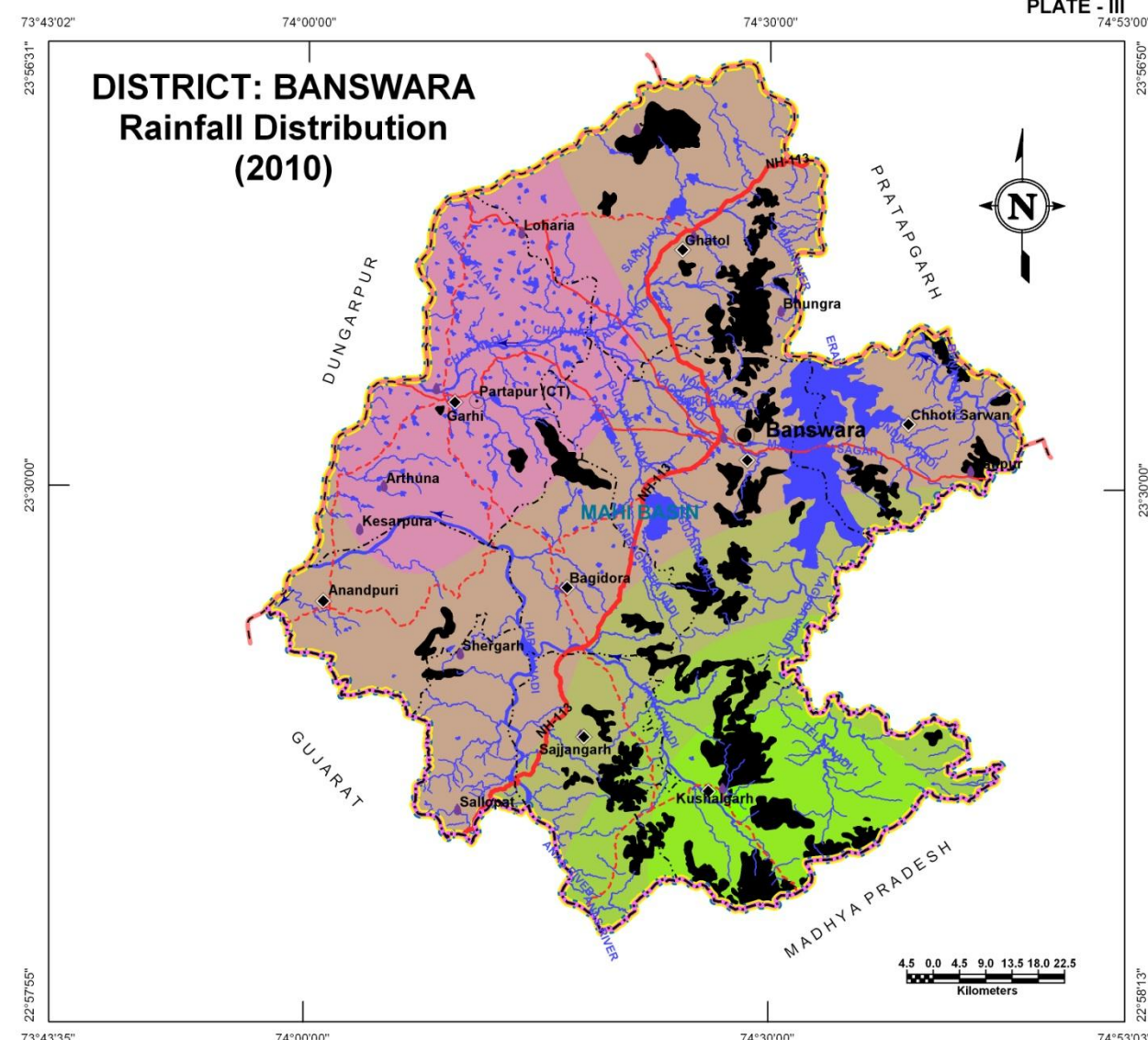
- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:
 - River / Streams
 - Ponds / Reservoirs

Elevation (m amsl):



Source : SRTM DEM

PLATE - III



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- Raingauge Stations
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Roads:
 - National Highway
 - State Highway
 - Major District Road

Water Bodies:

- River / Streams
- Ponds / Reservoirs
- Hills

Isohyets (mm):

- 500-600
- 600-700
- 700-800
- 800-900
- 900-1000
- > 1000

GEOLOGY

DISTRICT – BANSWARA

District geology includes Bhilwara Super Group, Aravalli meta-sediments, post-Aravalli intrusive and Younger flows. The Aravalli Super Group of formation consists in parts of Ghatol, Garhi, Banswara, Aandpuri and Sajjangarh blocks. The meta-sediments are comprised of quartzite, mica schist, conglomerates, quartz-chlorite schist, phyllites, dolomitic marble and gneisses. The post-Aravalli intrusive is granites, pegmatite, quartz veins and Amphibolite. The eastern and southeastern part covering Kushalgarh, part of Banswara and Chhoti Sarwan block, is occupied by Deccan trap basalt.

Super Group	Group	Formation
	Recent	Younger Alluvium and surface deposits (Clay and kankar with sand, gravel and rock debris), Grit and kankar
X-----X-----X-----X-----X-----X-----X---Unconformity---X-----X-----X-----X-----X-----X-----X-----		
	Deccan Traps	Basalt and intertrappean ash beds
Post Aravalli Intrusives		Granite, Pegmatites, Quartz veins, Amphibolite
Aravali		Composite gneiss phyllite, schist etc. with quartzite intercalations, limestone basalt, quartzite and local conglomerate. Intrusive quartz veins, pegmatite, granite aplite, talc-serpentine, chlorite schist amphibolite etc.
X-----X-----X-----X-----X-----X-----X---Unconformity---X-----X-----X-----X-----X-----X-----X-----		
Bhilwara	BGC	Gneisses, granite and Composite gneisses

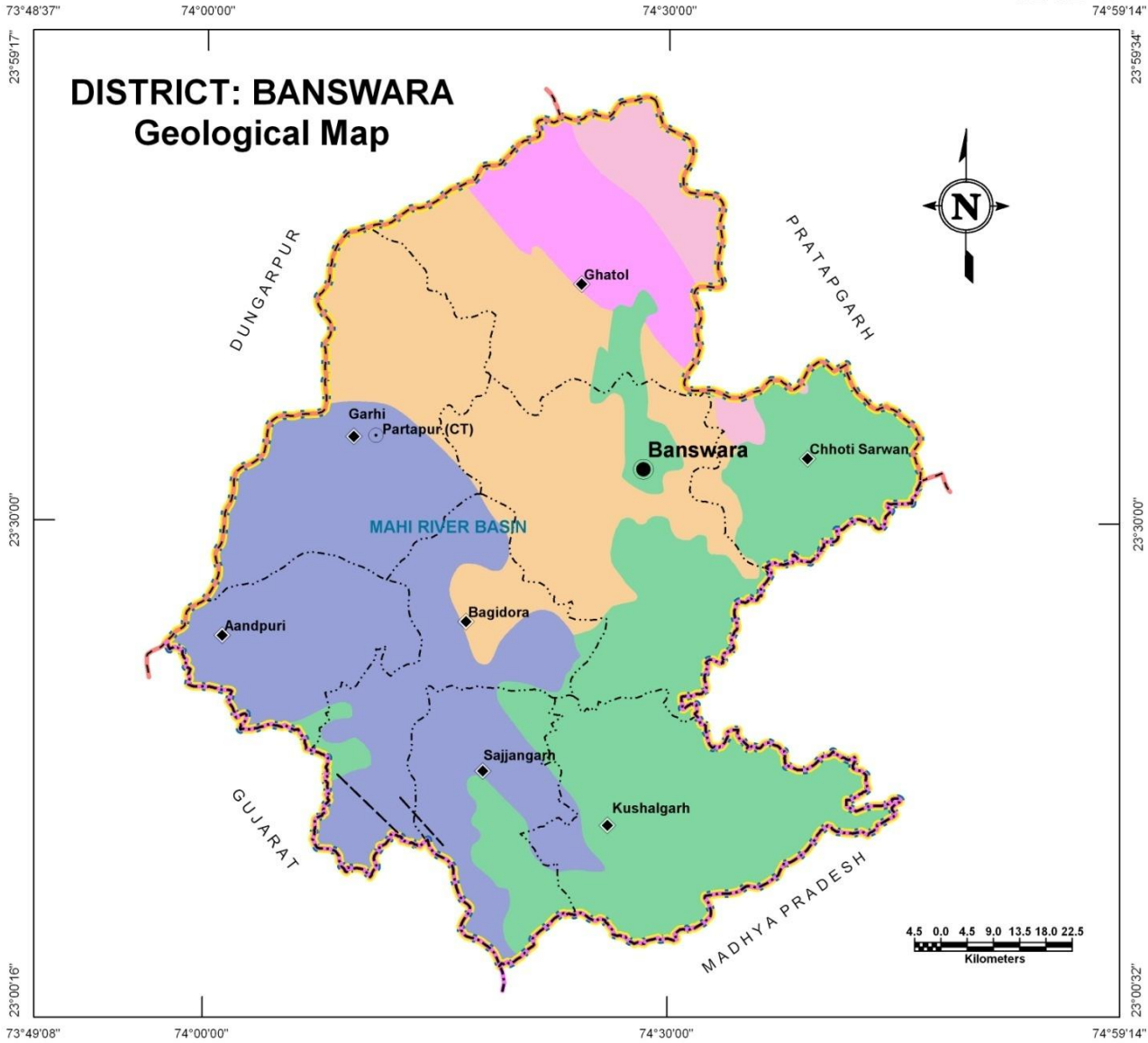
GEOMORPHOLOGY

Table: Geomorphologic units, their description and distribution

Origin	Landform Unit	Description
Denudational	Buried Pediment	Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.
	Intermontane Valley	Depression between mountains, generally broad & linear, filled with colluvial deposits.
	Pediplain	Coalescence and extensive occurrence of pediment.
	Pediment Zone	Formed by coalescence of several alluvial fans by stream covering large area at foot hills, with gentle slope in humid to sub humid region.
Fluvial	Valley Fill	Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.
Structural	Plateau	Formed over varying lithology with extensive, flat, landscapes, bordered by escarpment on all sides. Essentially formed horizontally layered rocky marked by extensive flat top and steep slopes. It may be criss crossed by lineament.
Hills	Denudational, Structural Hill, Linear Ridge	Steep sided, relict hills undergone denudation, comprising of varying lithology with joints, fractures and lineaments. Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc. Long narrow low-lying ridge usually barren, having high run off may form over varying lithology with controlled strike.



PLATE - IV



LEGEND

Admin Boundary:

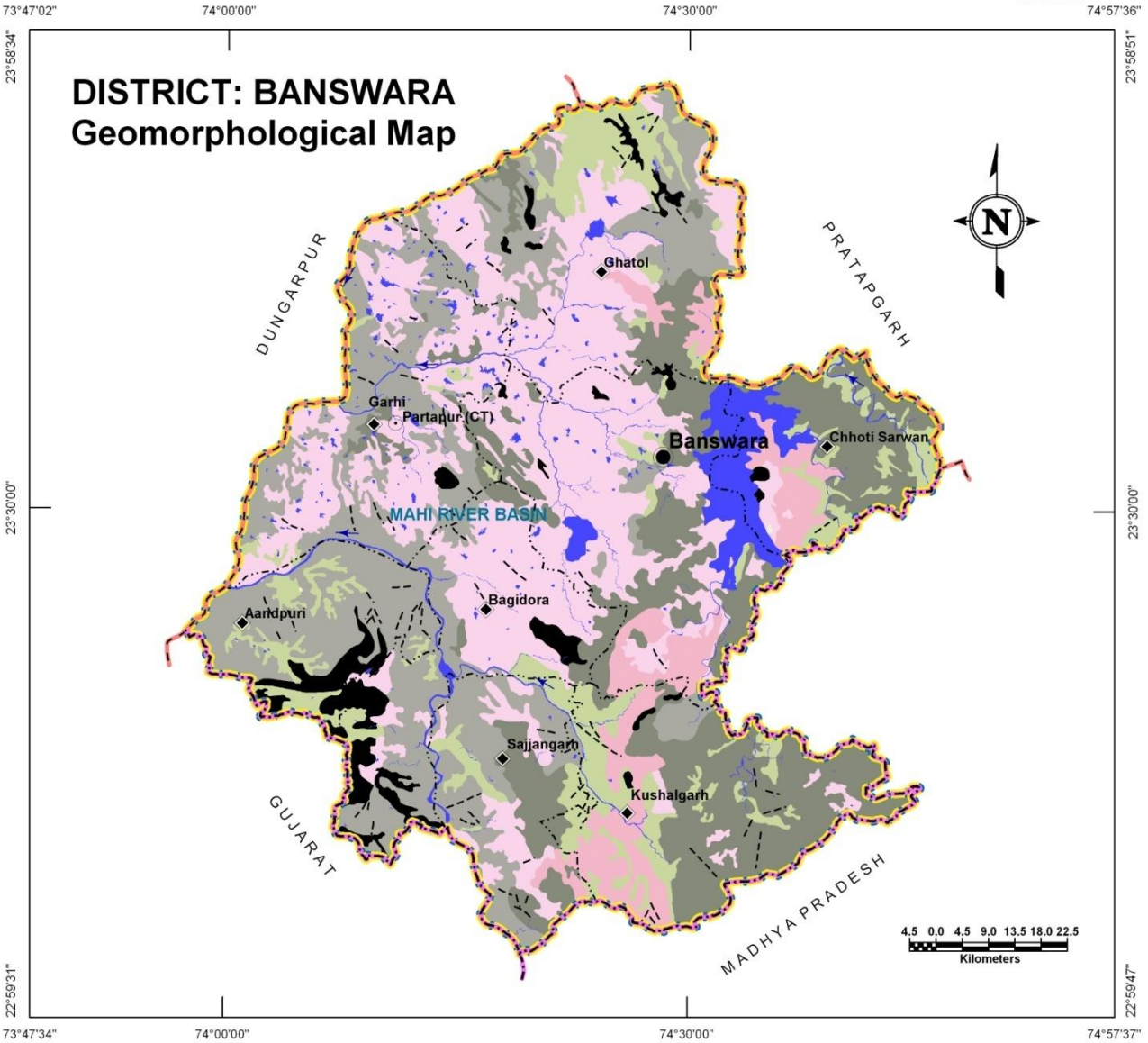
- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Structural Features:
- Fault

Geology:

- Udaipur Group
- Debari Group
- Mangalwar Complex
- Deccan Traps
- Lunavada Group

Source: District Resource Map of Rajasthan - GSI

PLATE - V



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:
- River/Ponds/Reservoirs
- Structural Features:
- Fault/Fractures/Lineament
- Hills:
- Structural/Denudational/Linear Ridge

Landform Units:

- Fluvial Origin:
- Denudational Origin:
- Structural Origin:

Source: Source: Ground Water Atlas of Rajasthan - SRSAC & GWD, Rajasthan

AQUIFERS

In Banswara district, aquifers are formed primarily in weathered, fractured and jointed hardrock formations of Phyllite, Basalt, BGC and Limestone in that order of spatial coverage. Phyllites aquifers occupy about 44% of district area, Basalt about 26% and BGC and limestone together constitute the rest of the aquifers excluding the hill part that has been excluded from aquifer delineation.

DISTRICT – BANSWARA

Table: aquifer potential zones their area and their description

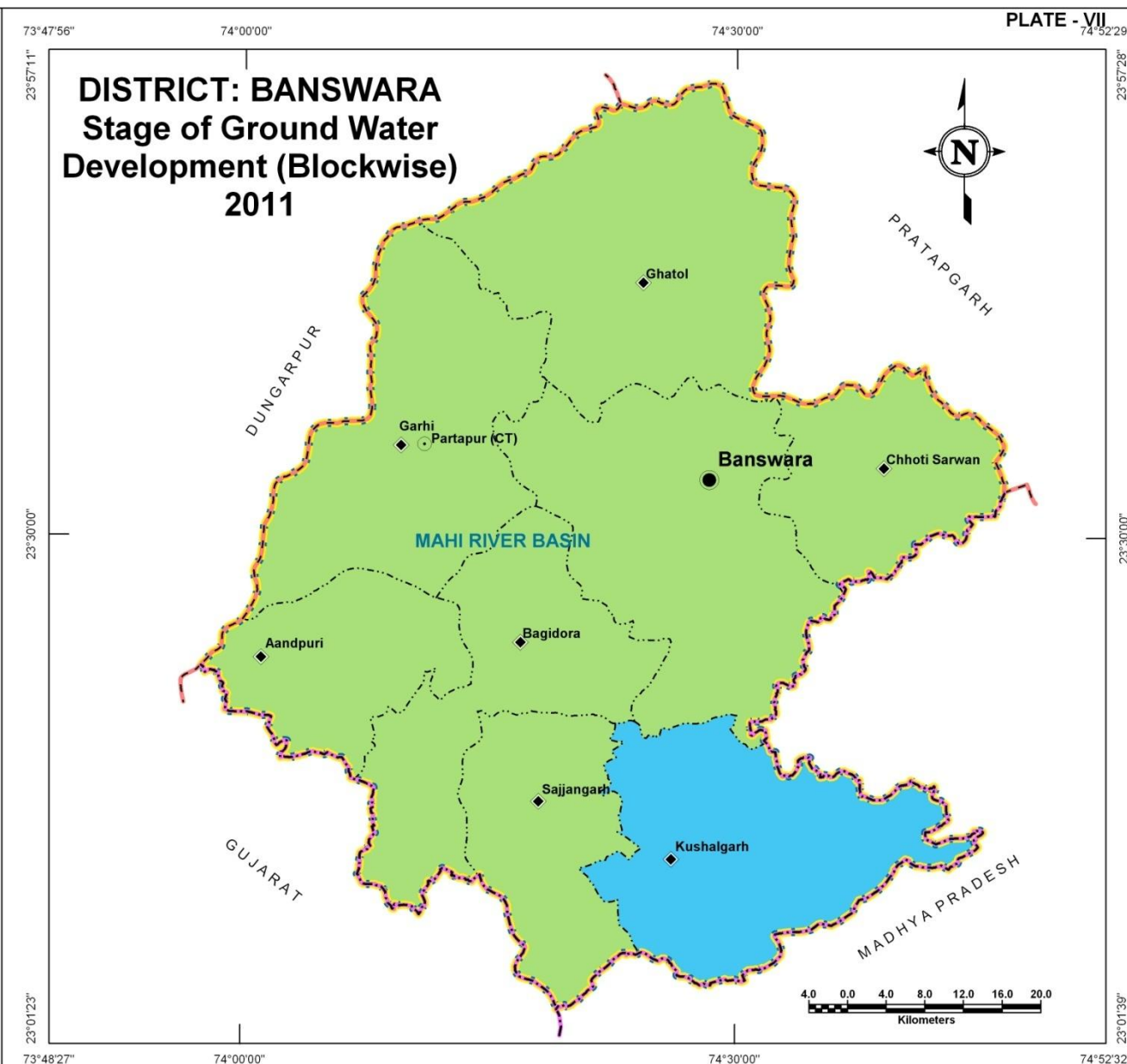
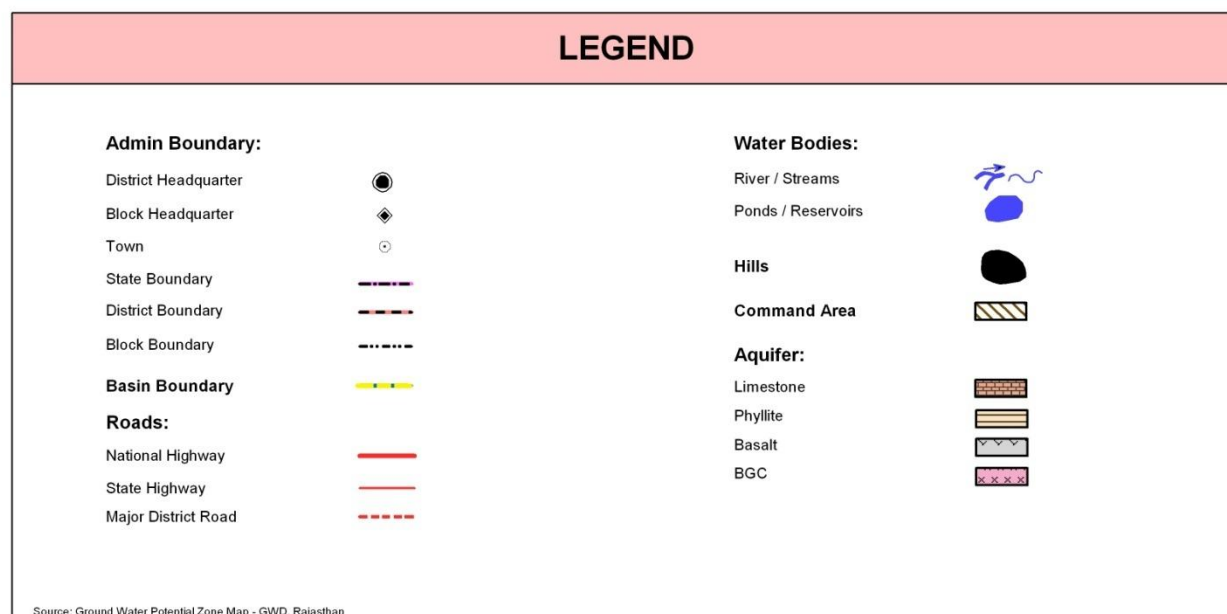
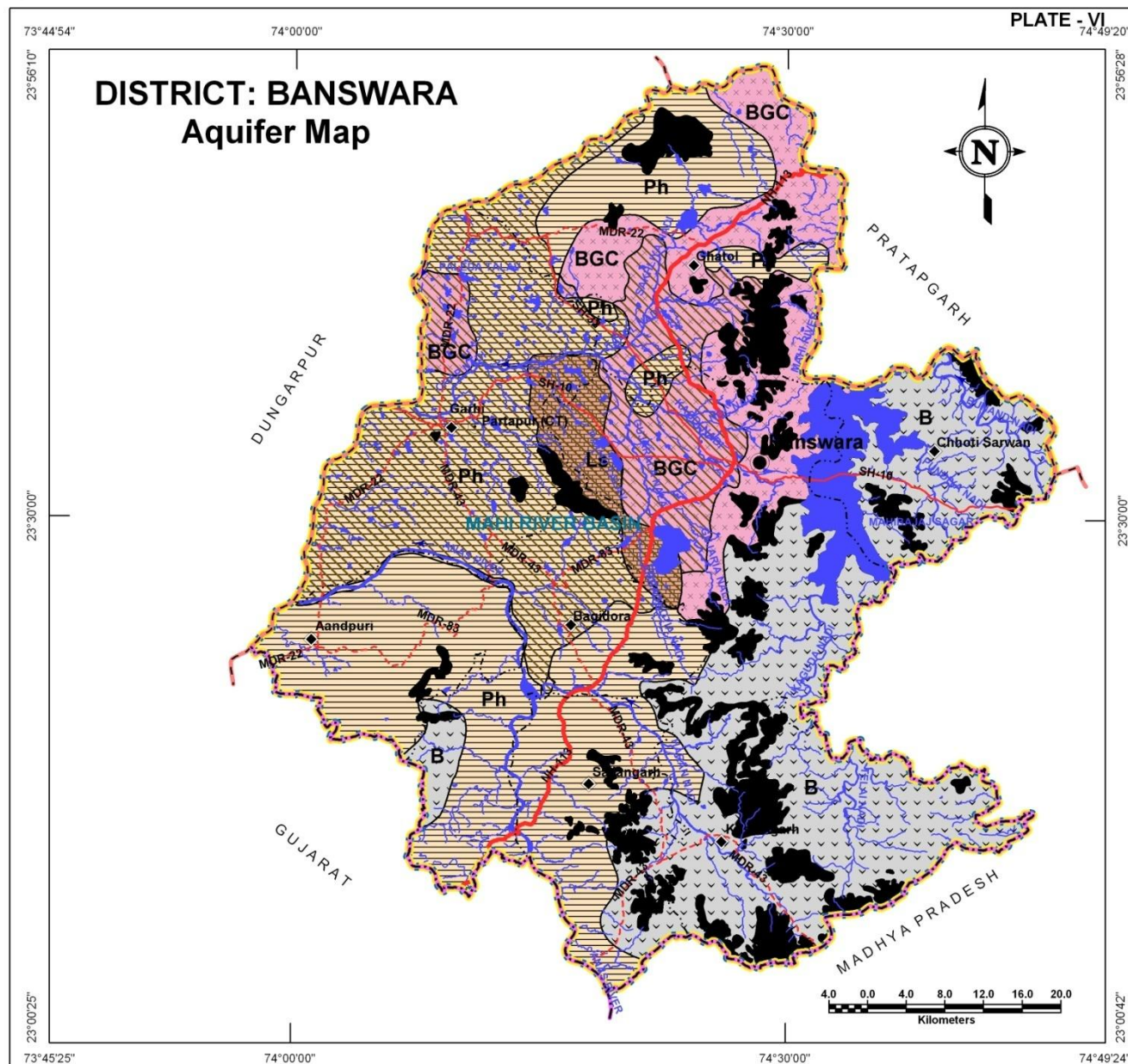
Aquifer in Potential Zone	Area (sq km)	% age of district	Description of the unit/Occurrence
Limestone	144.5	3.2	In general, it is fine to medium grained, grey, red yellowish, pink or buff in colour.
Phyllite	1,969.1	43.9	These include meta sediments and represented by carbonaceous phyllite.
Basalt	1,183.4	26.3	Dark grey, olive green and green colour, compact, vesicular, amygdaloidal and weathered.
BGC	776.5	17.3	Grey to dark coloured, medium to coarse grained rocks.
Non Potential Zone (Hills)	419.5	9.3	-
Total	4,493.0	100.0	

STAGE OF GROUND WATER DEVELOPMENT

Eight of the nine blocks fall under 'safe' category. This indicates that the ground water recharge has been more than the extraction and aquifers still have the possibility for future development. The only block that has been categorized as 'semi-critical' apparently has been developed fully.

Categorization on the basis of stage of development of ground water	Block Name
Safe	Ghatol, Chhoti Sarwan, Bagidora, Anandpuri, Sajjangarh, Banswara, Garhi
Semi-Critical	Kushalgarh

Basis for categorization: Ground water development <= 70% - Safe; 70 – 90% Semi critical.



LOCATION OF EXPLORATORY AND GROUND WATER MONITORING WELLS

DISTRICT – BANSWARA

The district has a well distributed network of large number of exploratory wells (91) and ground water monitoring stations (286) in the district owned by RGWD (59 and 256 respectively) and CGWB (32 and 30 respectively). The exploratory wells have formed the basis for delineation of subsurface aquifer distribution scenario in three dimensions. Benchmarking and optimization studies suggest that ground water level monitoring should be strengthened by adding 21 additional wells in four blocks and just 2 additional wells in Garhi block in order to effectively monitor the water quality in the district.

Table: Block wise count of wells (existing and recommended)

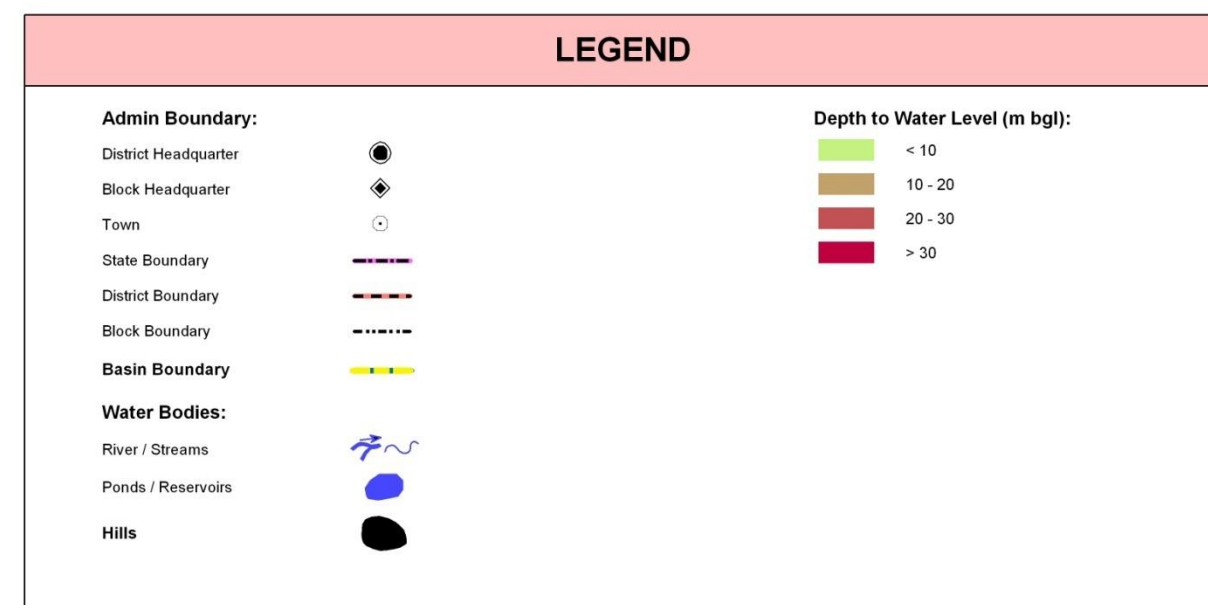
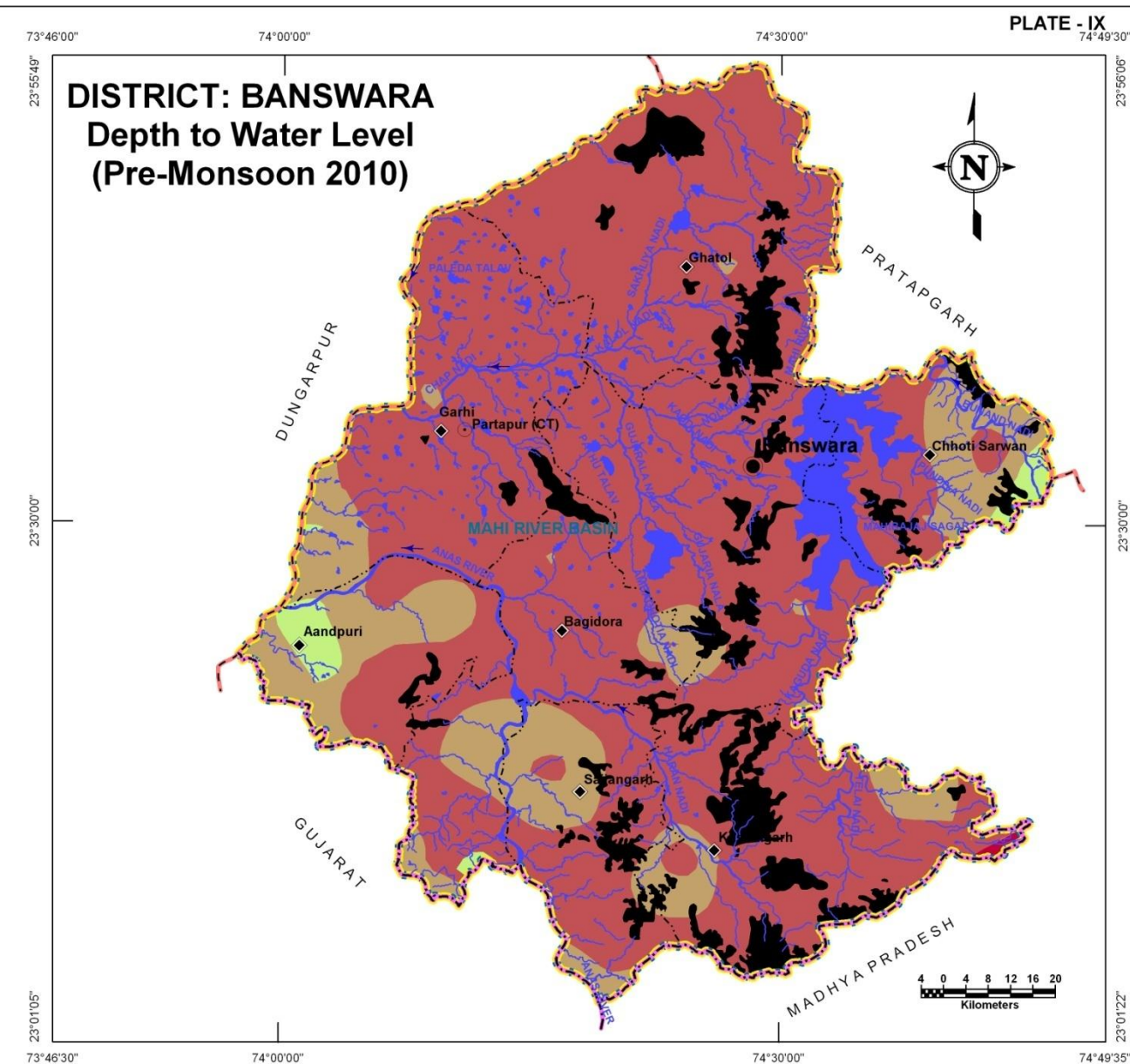
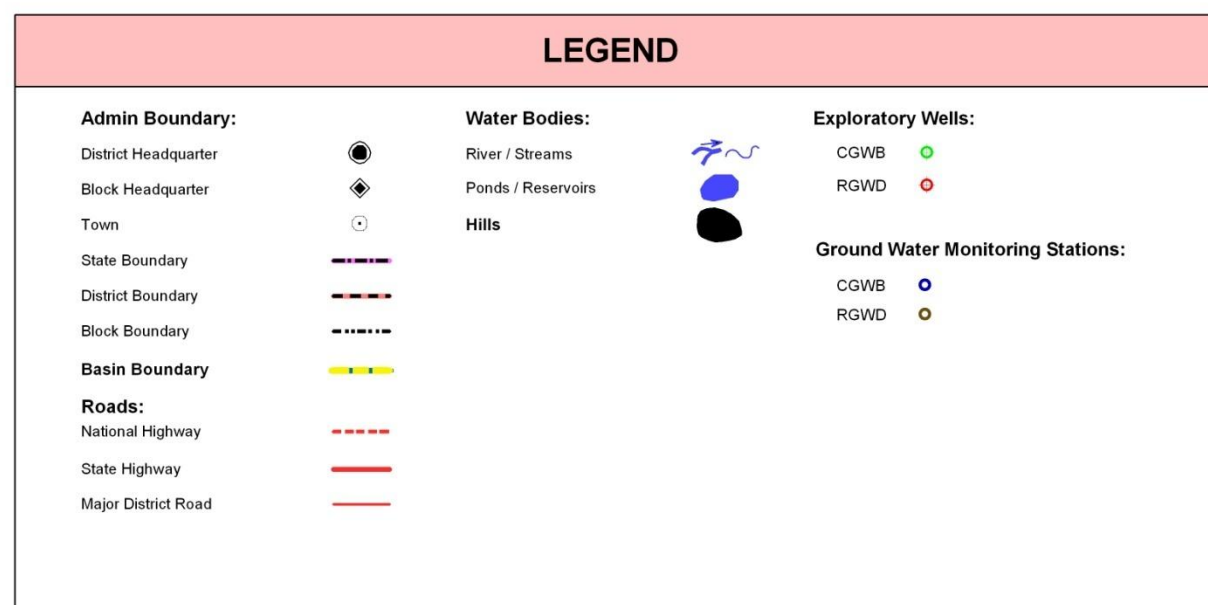
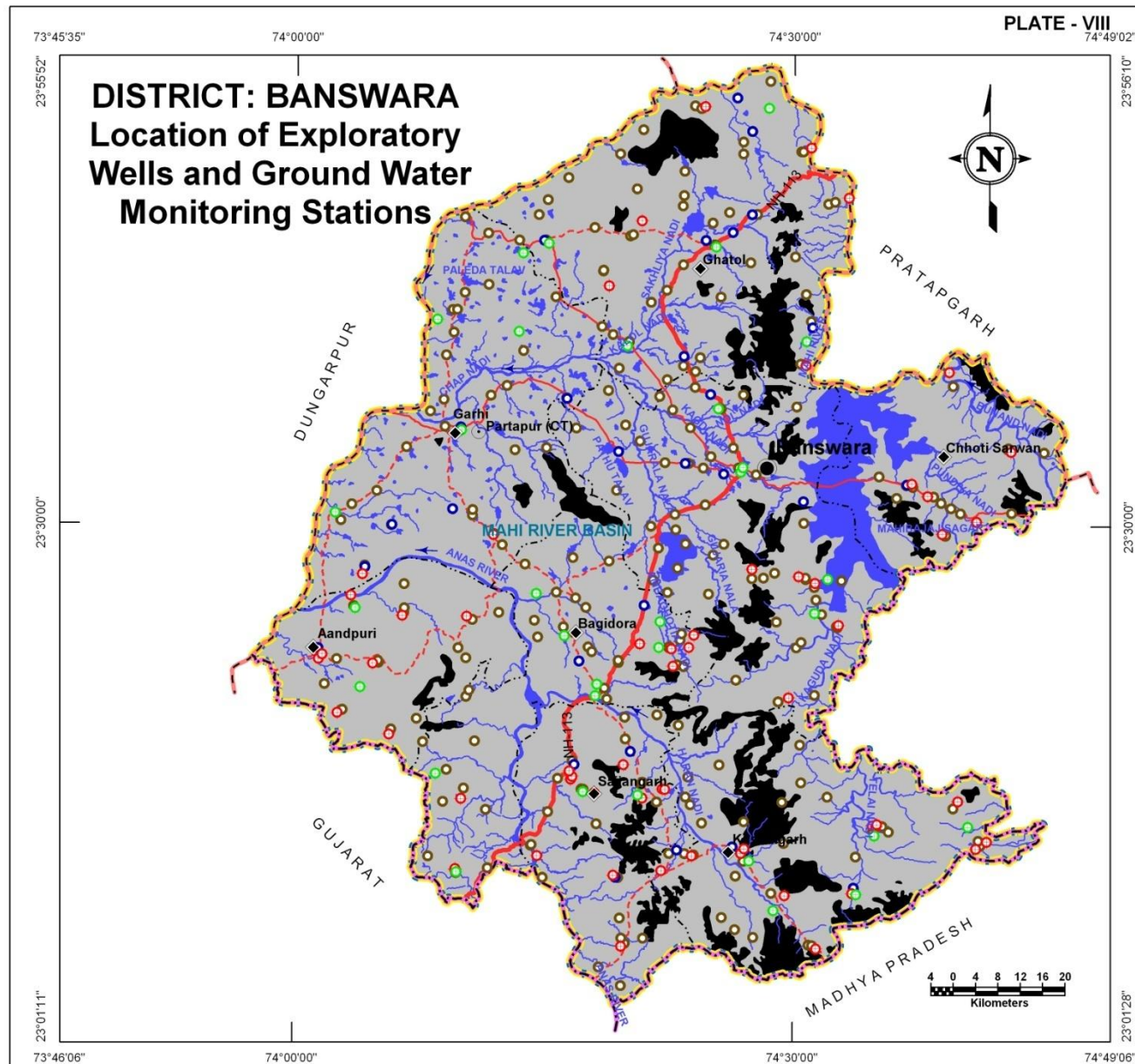
Block Name	Exploratory Wells			Ground Water Monitoring Stations			Recommended additional wells for optimization of monitoring network	
	CGWB	RGWD	Total	CGWB	RGWD	Total	Water Level	Water Quality
Anandpuri	2	8	10	1	19	20	-	-
Bagidora	8	8	16	2	32	34	-	-
Banswara	5	5	10	5	52	57	4	-
Chhoti Sarwan	-	6	6	1	15	16	5	-
Garhi	5	2	7	5	32	37	-	2
Ghatol	5	6	11	8	48	56	-	-
Kushalgarh	5	13	18	5	34	39	11	-
Sajjangarh	2	11	13	3	24	27	1	-
Total	32	59	91	30	256	286	21	2

DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

In spite of being a predominantly hard rock area, the district shows moderate variation in depth to ground water levels from less than 10m bgl to around 40m bgl. Shallow water levels of less than 10m bgl are seen in the southwestern parts of Anandpuri, Bagidora and Garhi blocks and northeast part of Chhoti Sarwan Block of the district. Deeper water levels of more than 30m bgl are found in Kushalgarh block covering a very small area (6.5 sq km). In rest of the district the depth of ground water occurrence varies between 10m to 30m bgl.

Depth to water level (m bgl)	Block wise area coverage (sq km) *								Total Area (sq km)
	Anandpuri	Bagidora	Banswara	Chhoti Sarwan	Garhi	Ghatol	Kushalgarh	Sajjangarh	
< 10	35.9	4.8	-	18.7	1.5	-	-	-	60.9
10-20	152.8	105.7	17.4	116.2	76.0	2.4	73.8	123.8	668.1
20-30	151.6	377.2	684.8	200.8	585.6	704.6	415.1	218.3	3,338.0
> 30	-	-	-	-	-	-	6.5	-	6.5
Total	340.3	487.7	702.2	335.7	663.1	707.0	495.4	342.1	4,073.5

* The area covered in the derived maps is less than the total district area since the hills have been excluded from interpolation/contouring.



WATER TABLE ELEVATION (PRE MONSOON – 2010)

DISTRICT – BANSWARA

Water table elevation map is presented in Plate – X. A perusal of the map reveals a general flow direction of ground water from east to west within in the district. In the southeastern part the water table is highest, reaching to > 400m amsl as seen in Kushalgarh and Chhoti Sarwan blocks of the district. The water table gradually lowers towards west reaching a minimum elevation (<120m amsl) in the Anandpuri and Garhi blocks. This thus shows a drop in water level within the district by more than 280m. The flow gradients are higher in the eastern parts of the district and sluggish in western parts.

Table: Block wise area covered in each water table elevation range

Water table elevation range (m amsl)	Block wise area coverage (sq km)								Total Area (sq km)
	Anandpuri	Bagidora	Banswara	Chhoti Sarwan	Garhi	Ghatol	Kushalgarh	Sajjangarh	
< 120	-	-	-	-	1.8	-	-	-	1.8
120 – 140	84.1	-	-	-	140.5	-	-	-	224.6
140 – 160	53.5	-	0.2	-	264.5	3.4	-	-	321.6
160 – 180	57.5	15.2	67.9	-	226.5	165.8	-	-	532.9
180 – 200	64.6	88.1	132.9	-	29.8	177.3	-	-	492.7
200 – 220	47.7	163.8	98.0	-	-	209.0	-	121.2	639.7
220 – 240	25.5	154.2	73.2	-	-	99.6	-	109.4	461.9
240 – 260	7.4	52.9	66.0	1.7	-	51.9	65.3	87.0	332.2
260 – 280	-	7.7	38.5	43.5	-	-	40.7	19.6	150.0
280 – 300	-	4.0	83.8	64.1	-	-	42.6	4.9	199.4
300 – 320	-	1.8	57.2	66.8	-	-	35.8	-	161.6
320 – 340	-	-	38.6	72.0	-	-	19.7	-	130.3
340 – 360	-	-	19.9	45.1	-	-	50.9	-	115.9
360 – 380	-	-	21.5	18.4	-	-	61.4	-	101.3
380 – 400	-	-	4.5	13.3	-	-	153.8	-	171.6
> 400	-	-	-	10.8	-	-	25.2	-	36.0
Total	340.3	487.7	702.2	335.7	663.1	707.0	495.4	342.1	4,073.5

WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

Water level fluctuation map (Plate – XI) shows a maximum rise of 16m is seen in the district in the eastern part (Kushalgarh block). The 2m contour interval maps shows ground water level fluctuation from 0 m in one area to rise in other areas reaching upto significant levels. The central and northern parts of the district show a general rise in the water levels by 0 to 6m in the post-monsoon period when compared to pre-monsoon season water levels. Overall, the –ve fluctuation areas are very limited in spatial distribution and occupy very small area of 6.4 sq kms in the northernmost part of the district in Ghatol block.

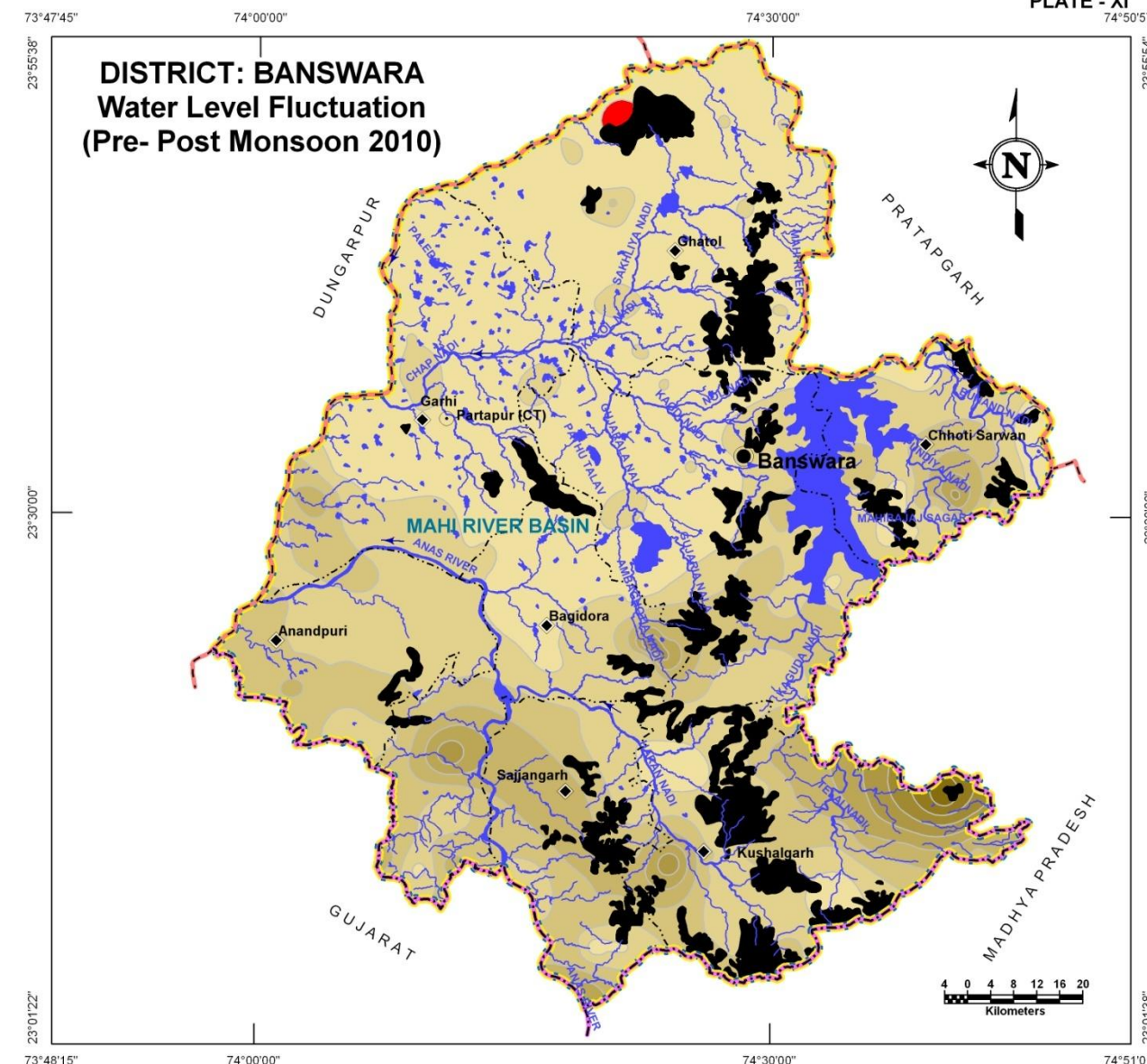
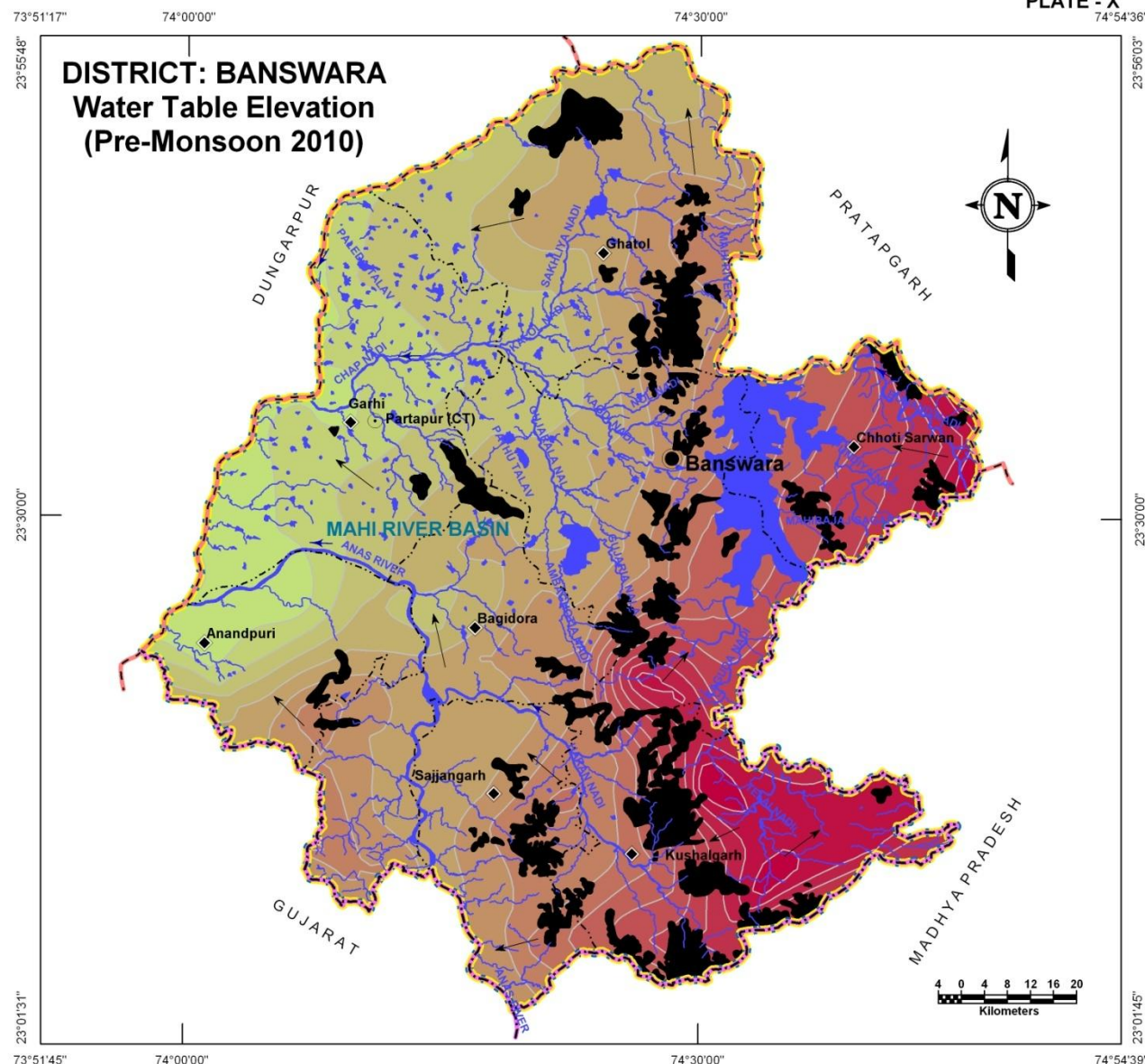
Table: Block wise area covered in each water fluctuation zone

Water level fluctuation range (m)	Block wise area coverage (sq km)								Total Area (sq km)
	Anandpuri	Bagidora	Banswara	Chhoti Sarwan	Garhi	Ghatol	Kushalgarh	Sajjangarh	
<0	-	-	-	-	-	6.4	-	-	6.4
0 – 2	0.2	145.5	332.7	55.6	476.3	662.3	46.1	6.0	1,724.7
2 – 4	209.5	140.9	250.0	184.2	129.2	37.8	143.4	94.1	1,189.1
4 – 6	127.3	114.8	115.6	68.9	49.5	0.5	112.8	132.2	721.6
6 – 8	3.3	50.8	3.9	19.7	8.1	-	85.4	103.9	275.1
8 – 10	-	27.4	-	6.7	-	-	45.1	5.9	85.1
10 – 12	-	8.3	-	0.6	-	-	29.1	-	38.0
12 – 14	-	-	-	-	-	-	17.8	-	17.8
14 – 16	-	-	-	-	-	-	10.9	-	10.9
>16	-	-	-	-	-	-	4.8	-	4.8
Total	340.3	487.7	702.2	335.7	663.1	707.0	495.4	342.1	4,073.5



PLATE - X

PLATE - XI



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:
- River / Streams
- Ponds / Reservoirs
- Hills
- Water Table Elevation

Water Table Elevation(m amsl):

- <120
- 120 - 140
- 140 - 160
- 160 - 180
- 180 - 200
- 200 - 220
- 220 - 240
- 240 - 260
- 260 - 280
- 280 - 300
- 300 - 320
- 320 - 340
- 340 - 360
- 360 - 380
- 380 - 400
- >400

LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:
- River / Streams
- Ponds / Reservoirs
- Hills

Water Level Fluctuation(m):

- <0
- 0 to 2
- 2 to 4
- 4 to 6
- 6 to 8
- 8 to 10
- 10 to 12
- 12 to 14
- 14 to 16
- >16

GROUND WATER ELECTRICAL CONDUCTIVITY DISTRIBUTION

DISTRICT – BANSWARA

The Electrical conductivity (at 25°C) distribution map is presented in Plate – XII. The areas with low EC values in ground water (<2000 µS/cm) are shown in yellow color and it is interesting to find that the whole district is falling within the same EC range implying that the ground water data of the whole district is suitable for domestic purpose.

Table: Block wise area of Electrical conductivity distribution

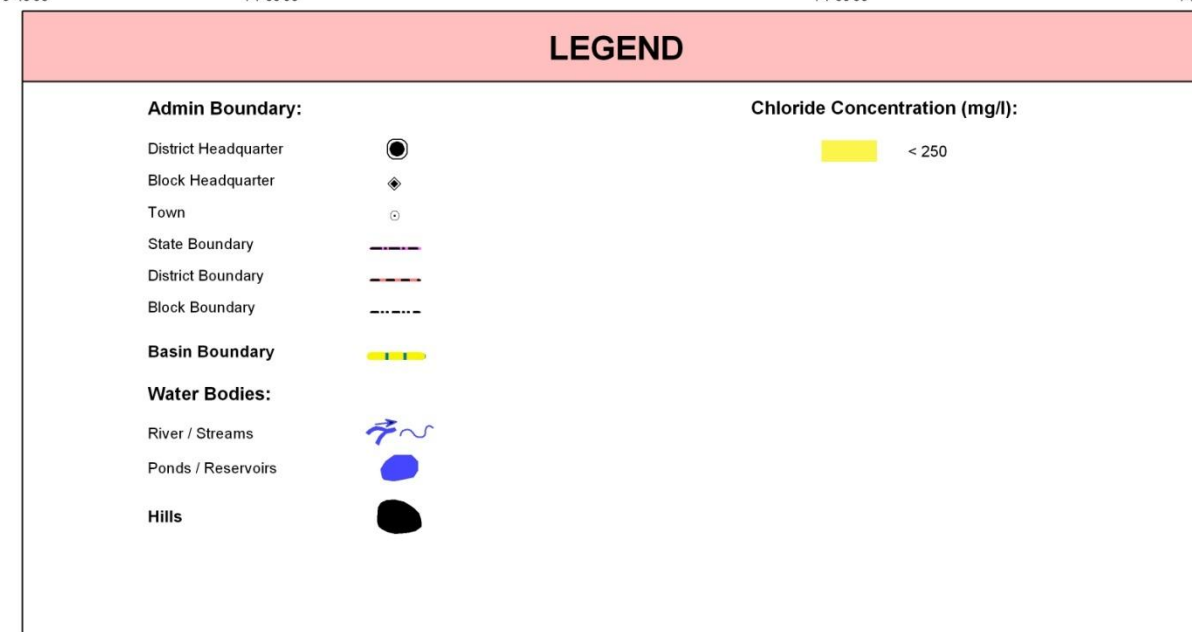
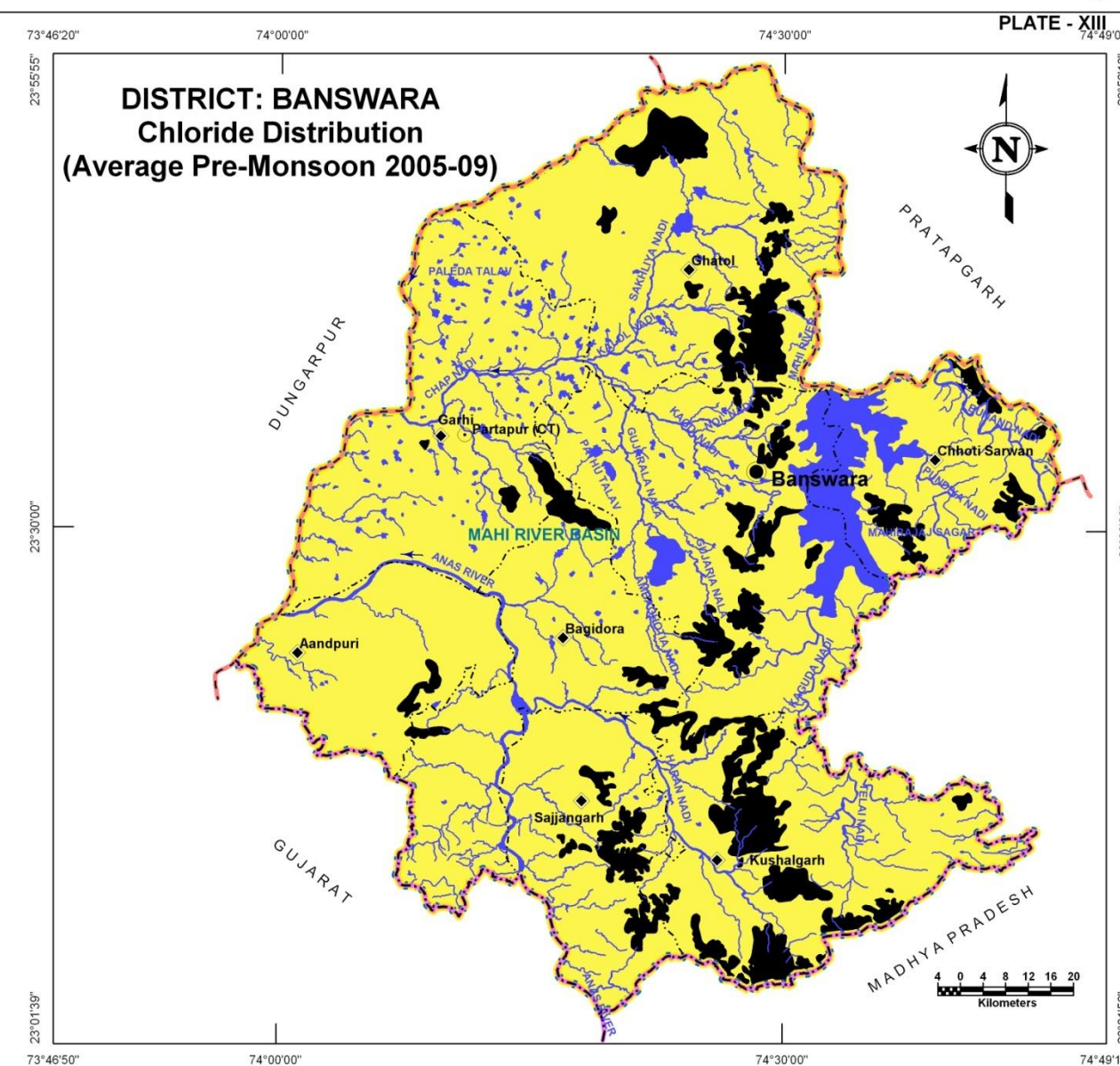
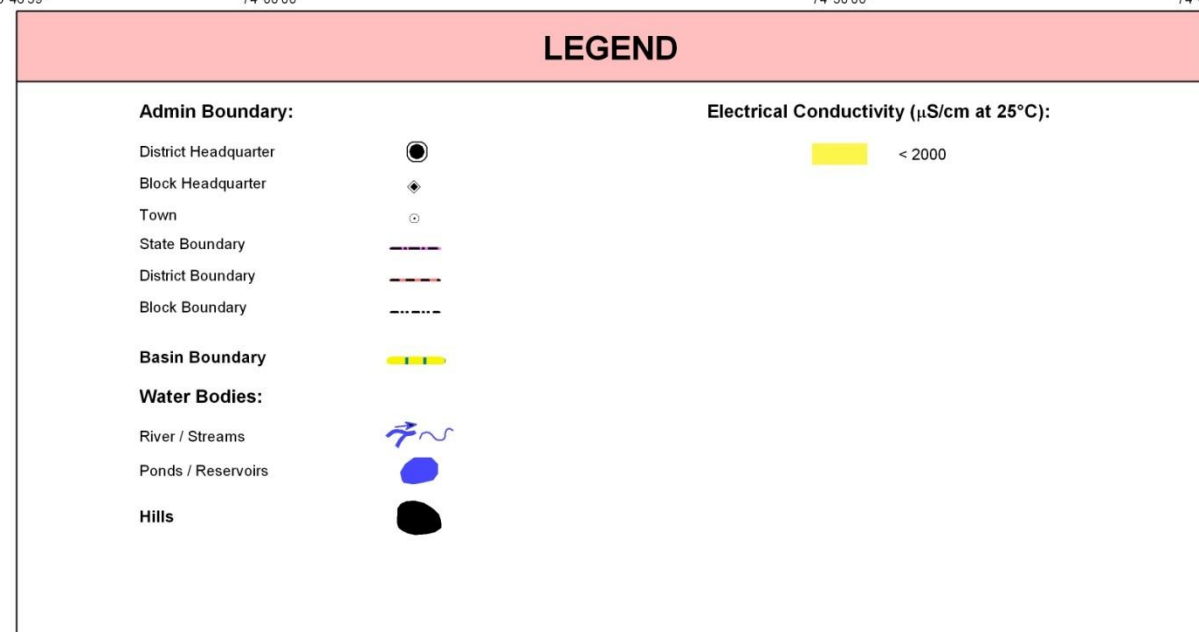
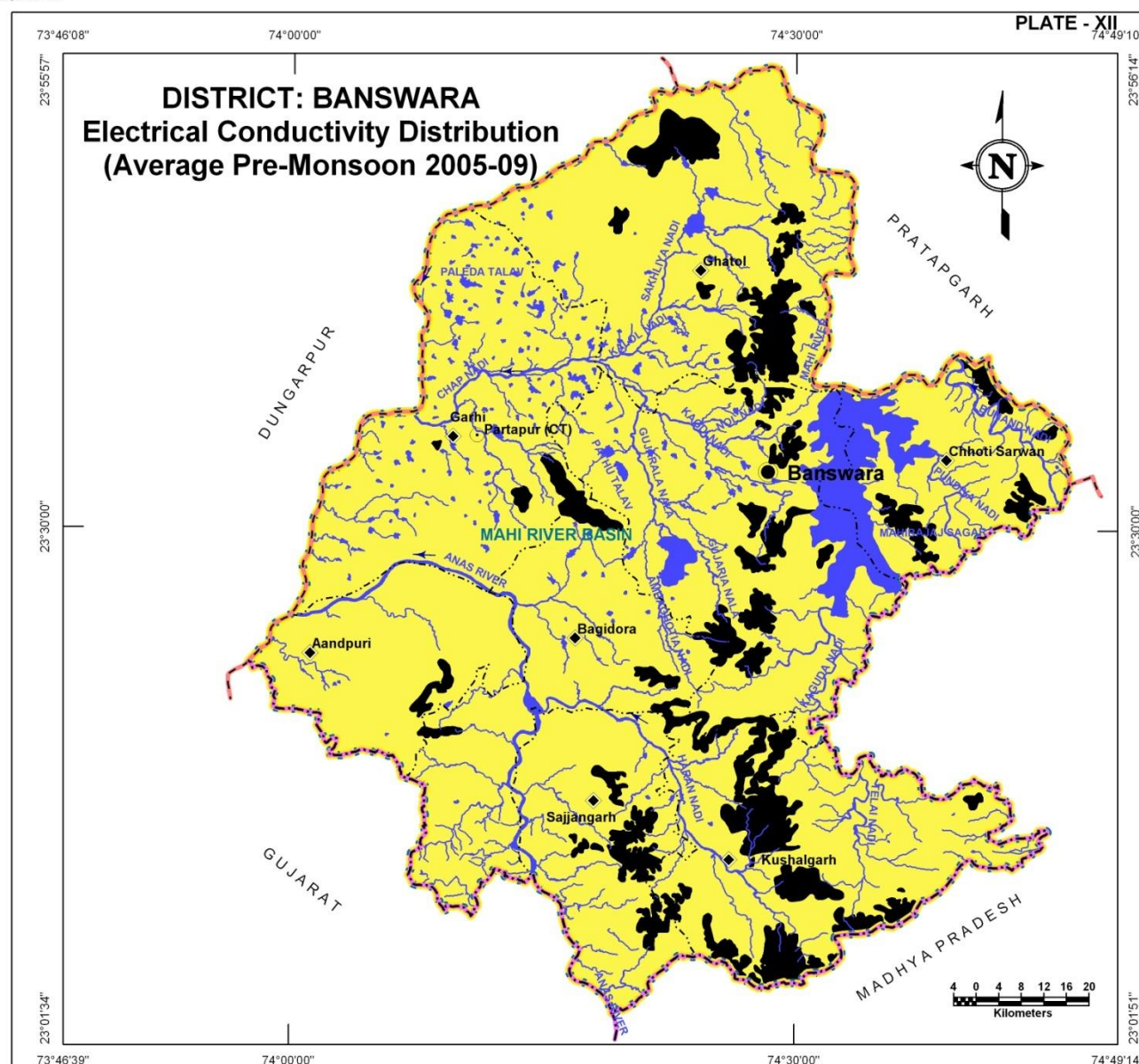
Electrical Conductivity Ranges (µS/cm at 25°C) (Ave. of years 2005-09)	Block wise area coverage (sq km)																Total Area (sq km)
	Anandpuri		Bagidora		Banswara		Chhoti Sarwan		Garhi		Ghatol		Kushalgarh		Sajjangarh		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 2000	340.3	100.0	487.7	100.0	702.2	100.0	335.7	100.0	663.1	100.0	707.0	100.0	495.4	100.0	342.1	100.0	4,073.5
2000-4000	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
>4000	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total	340.3	100.0	487.7	100.0	702.2	100.0	335.7	100.0	663.1	100.0	707.0	100.0	495.4	100.0	342.1	100.0	4,073.5

GROUND WATER CHLORIDE DISTRIBUTION

The chloride concentration map is presented in Plate – XIII which reveals the same pattern of distribution of concentration areas as that of EC as above. The whole district has shown low chloride concentration in ground water (<250 mg/l) implying that there are no areas where the water is unsuitable for domestic purposes.

Table: Block wise area of Chloride distribution

Chloride Concentration Range (mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)																Total Are (sq km)
	Anandpuri		Bagidora		Banswara		Chhoti Sarwan		Garhi		Ghatol		Kushalgarh		Sajjangarh		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 250	340.3	100.0	487.7	100.0	702.2	100.0	335.7	100.0	663.1	100.0	707.0	100.0	495.4	100.0	342.1	100.0	4,073.
250-1000	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
> 1000	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Total	340.3	100.0	487.7	100.0	702.2	100.0	335.7	100.0	663.1	100.0	707.0	100.0	495.4	100.0	342.1	100.0	4,073.



GROUND WATER FLUORIDE DISTRIBUTION

DISTRICT – BANSWARA

The Fluoride concentration map is presented in Plate XIV. The areas with low concentration (i.e., >1.5 mg/l) are shown in yellow color and occupies almost 95% of the district area which is suitable for domestic purpose. The areas with moderately high concentration (1.5-3.0 mg/l) are shown in green color small patches scattered randomly over the district. Remaining small part of district falls under high fluoride concentration (>3.0 mg/l) which is shown in red color northern part of the district near Ghatol where the ground water is not suitable for domestic purposes.

Table: Block wise area of Fluoride distribution

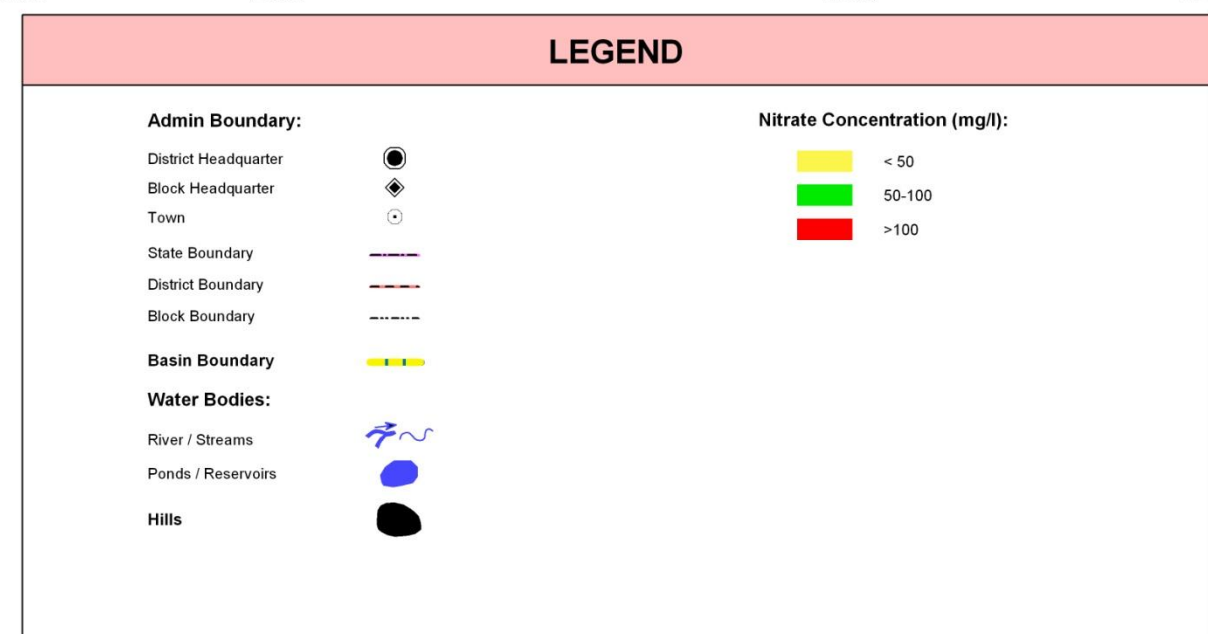
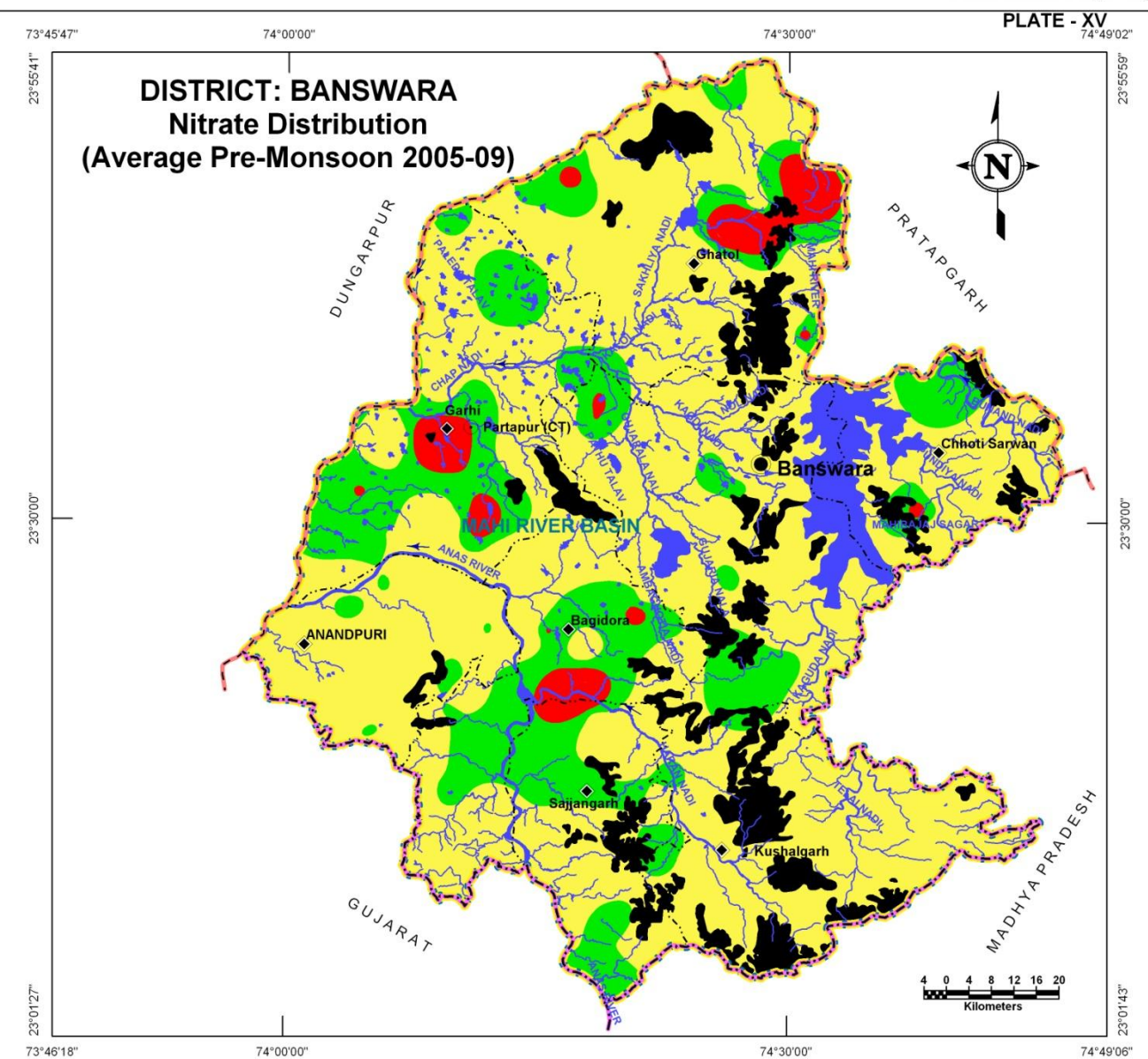
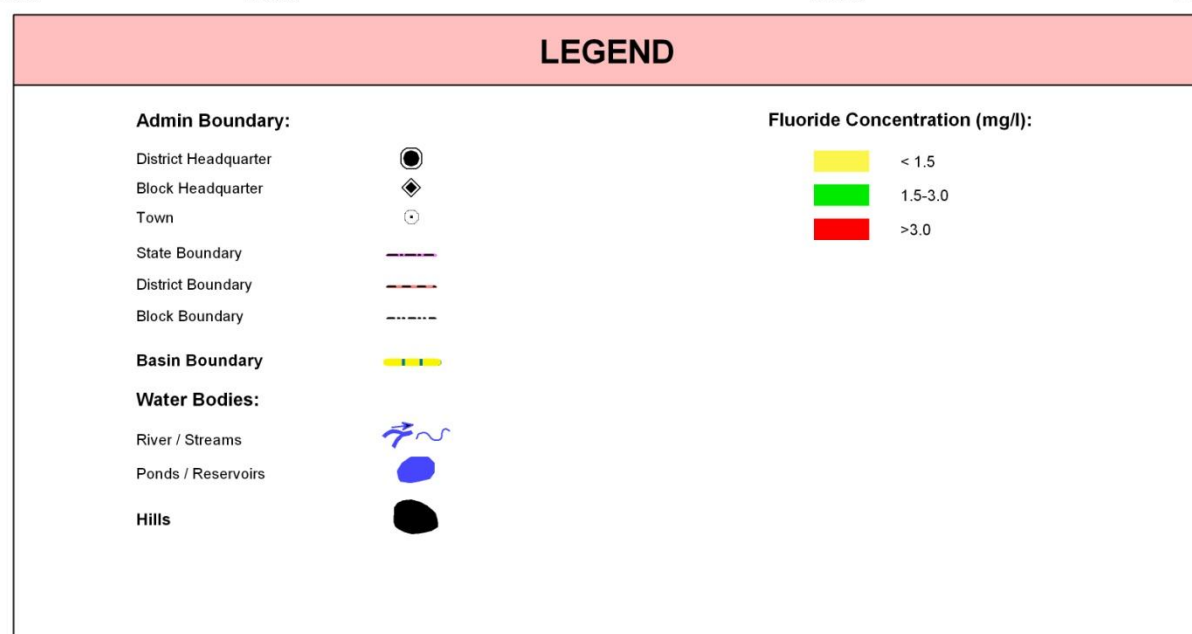
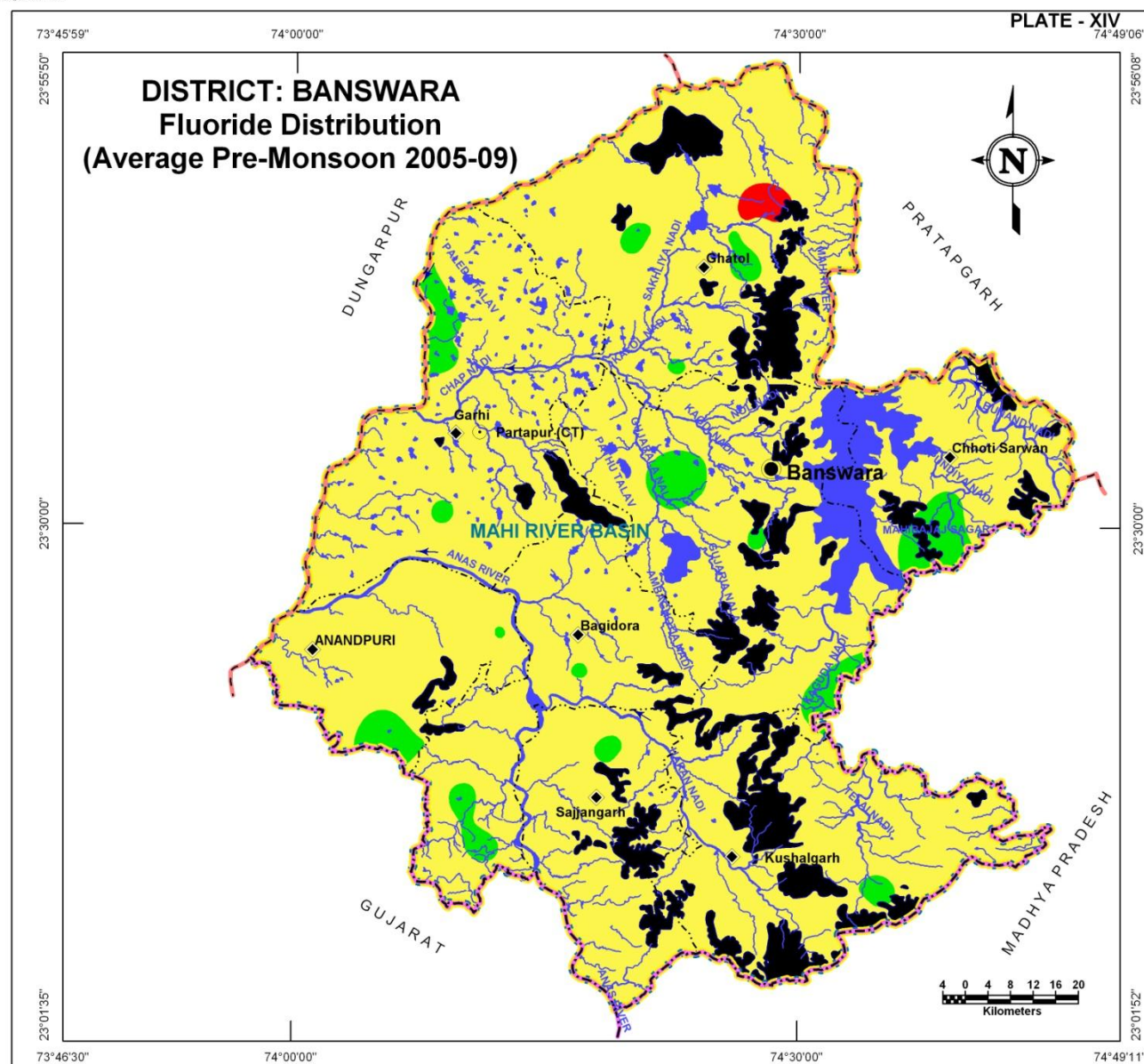
Fluoride concentration range (mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)																Total Area (sq km)
	Anandpuri		Bagidora		Banswara		Chhoti Sarwan		Garhi		Ghatol		Kushalgarh		Sajjangarh		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 1.5	316.4	93.0	466.3	95.6	647.2	92.2	298.9	89.0	626.7	94.5	671.6	95.0	487.3	98.4	336.5	98.4	3,850.9
1.5-3.0	23.9	7.0	21.4	4.4	55.0	7.8	36.8	11.0	36.4	5.5	20.1	2.8	8.1	1.6	5.6	1.6	207.3
> 3.0	-	-	-	-	-	-	-	-	-	-	15.3	2.2	-	-	-	-	15.3
Total	340.3	100.0	487.7	100.0	702.2	100.0	335.7	100.0	663.1	100.0	707.0	100.0	495.4	100.0	342.1	100.0	4,073.5

GROUND WATER NITRATE DISTRIBUTION

Nitrate concentration distribution maps helps in evaluating the ground water's suitability for agriculture purposes. Plate – XV shows distribution of Nitrate in ground water. Low concentration area of nitrate (<50 mg/l) is shown in yellow color and occupies approximately 75% of the district area which is suitable for agriculture purpose. The areas with moderately high nitrate concentration (50-100 mg/l) are shown in green color and occupy approximately 21% of the district area and except for southeastern part; such areas are distributed as pockets all over the district. In remaining part of the district the ground water has high nitrate concentration (>100 mg/l) as seen in red colored patches scattered, largely northern and western parts of the district where ground water is not suitable for agriculture purpose.

Table: Block wise area of Nitrate distribution

Nitrate concentration range (mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)																Total Area (sq km)
	Anandpuri		Bagidora		Banswara		Chhoti Sarwan		Garhi		Ghatol		Kushalgarh		Sajjangarh		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 50	324.3	95.3	291.0	59.6	602.1	85.7	277.7	82.7	361.7	54.6	519.8	73.5	475.9	96.1	196.5	57.4	3,049.0
50-100	16.0	4.7	170.5	35.0	97.4	13.9	56.2	16.8	258.7	39.0	119.7	16.9	19.5	3.9	135.3	39.6	873.0
>100	-	-	26.2	5.4	2.7	0.4	1.8	0.5	42.7	6.4	67.5	9.6	-	-	10.3	3.0	151.0
Total	340.3	100.0	487.7	100.0	702.2	100.0	335.7	100.0	663.1	100.0	707.0	100.0	495.4	100.0	342.1	100.0	4,073.0



DEPTH TO BEDROCK

DISTRICT – BANSWARA

From hydrogeological perspective, the beginning of massive bedrock has been considered for defining top of bedrock surface. The major rocks types found in the district are Limestone, Phyllite, Basalt and BGC. On perusal of the map (Plate – XVII) it can be interpreted that the bedrock is encountered at shallow depths in western half of the district whereas on moving to northern and southeastern side it increases reaching to depth of about 120m bgl. The central part (in Anandpuri, Bagidora, Banswara, Chhoti Sarwan and Sajjangarh blocks) and southern part (in Garhi, Ghatol and Kushalgarh blocks) of the district depth to bedrock varies with a thickness range of 20 – 80 meter below ground level. Isolate patches of more than 100 meter thickness found in Ghatol and Kushalgarh blocks.

Depth to bedrock (m bgl)	Block wise area coverage (sq km)																Total Area (sq km)
	Anandpuri		Bagidora		Banswara		Chhoti Sarwan		Garhi		Ghatol		Kushalgarh		Sajjangarh		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 20	15.3	5.0	1.3	-	-	-	-	-	10.8	2.0	5.9	1.0	-	-	57.2	17.0	90.5
20 - 40	83.2	24.0	75.5	15.0	6.9	1.0	-	-	289.9	43.0	118.4	17.0	34.2	7.0	128.7	38.0	736.8
40 - 60	145.3	42.0	168.3	34.0	535.3	76.2	280.9	83.7	169.9	26.0	319.4	45.0	188.1	38.0	155.0	45.0	1,962.2
60 - 80	66.5	20.0	179.0	38.0	135.0	19.2	54.8	16.3	128.0	19.0	186.8	26.0	213.1	43.0	1.2	-	964.4
80 - 100	30.0	9.0	63.6	13.0	25.0	3.6	-	-	64.5	10.0	44.4	6.0	49.8	10.0	-	-	277.3
100 - 120	-	-	-	-	-	-	-	-	-	-	32.1	5.0	8.4	2.0	-	-	40.5
> 120	-	-	-	-	-	-	-	-	-	-	-	-	1.8	-	-	-	1.8
Total	340.3	100.0	487.7	100.0	702.2	100.0	335.7	100.0	663.1	100.0	707.0	100.0	495.4	100.0	342.1	100.0	4,073.5

UNCONFINED AQUIFER

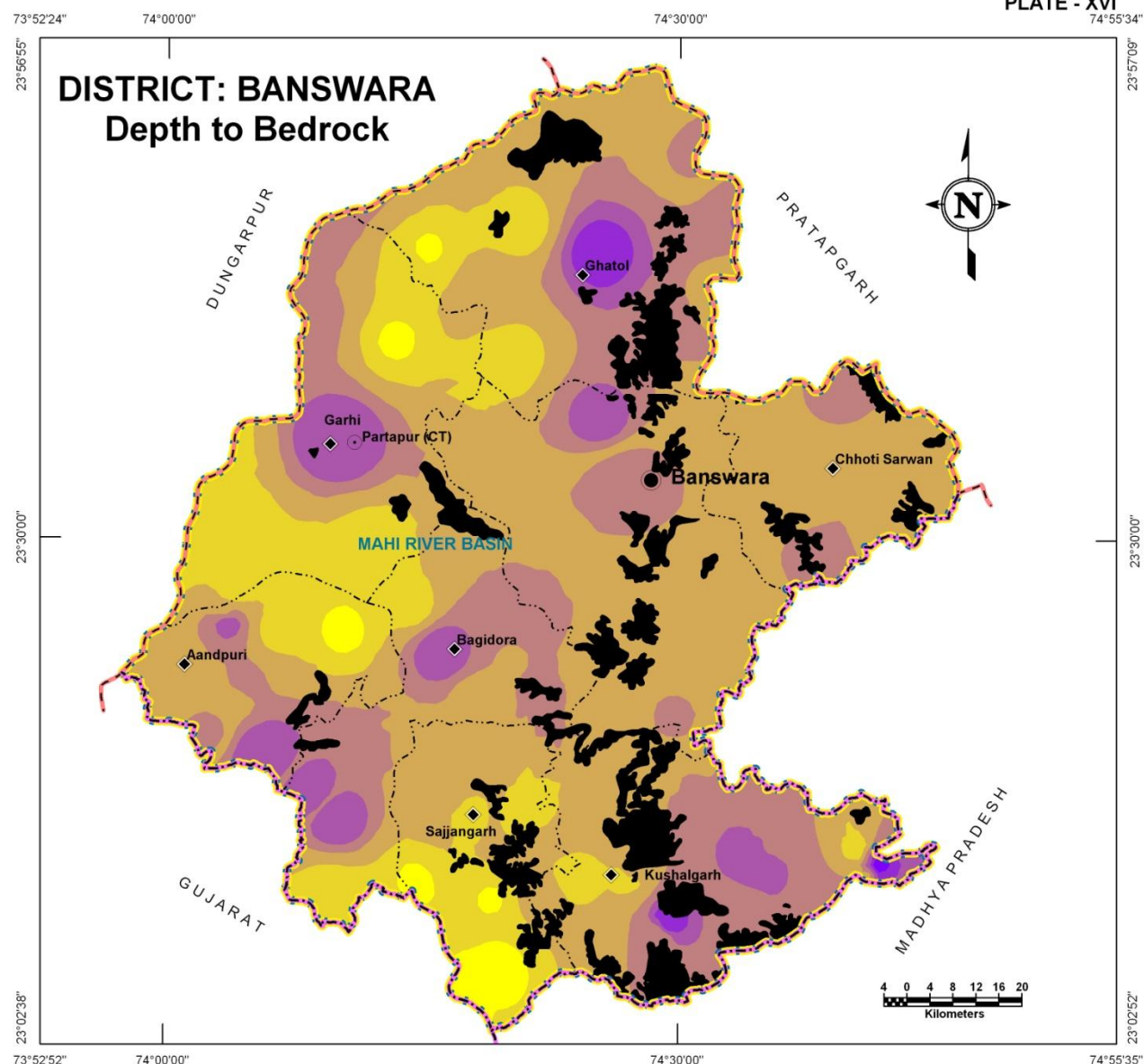
Hardrock areas

Aquifers in the district are predominantly formed in weathered, fractured and jointed rock formations occurring at shallower depths and these constitute good unconfined aquifers. The thickness of aquifers thus formed varies from less than 10m to around 60m. The southwestern part of Anandpuri block and northeastern part of Kushalgarh has very thick zones (> 50m) but less in terms of spatial coverage. Major areas of the district have shown the general thickness of upto 30m or less.

Unconfined aquifer Thickness (m)	Block wise area coverage (sq km)								Total Area (sq km)
	Anandpuri	Bagidora	Banswara	Chhoti Sarwan	Garhi	Ghatol	Kushalgarh	Sajjangarh	
< 10	109.3	92.5	154.5	87.2	271.6	416.7	211.3	299.6	1,642.7
10-20	103.0	211.6	313.2	190.1	239.1	271.7	216.1	41.1	1,585.9
20-30	77.0	136.4	143.4	0.3	127.9	16.6	47.5	1.4	550.5
30-40	39.1	47.2	84.7	49.8	24.5	2.0	5.0	-	252.3
40-50	11.4	-	6.4	8.3	-	-	2.0	-	28.1
> 50	0.5	-	-	-	-	-	13.5	-	14.0
Total	340.3	487.7	702.2	335.7	663.1	707	495.4	342.1	4,073.5



PLATE - XVI



LEGEND

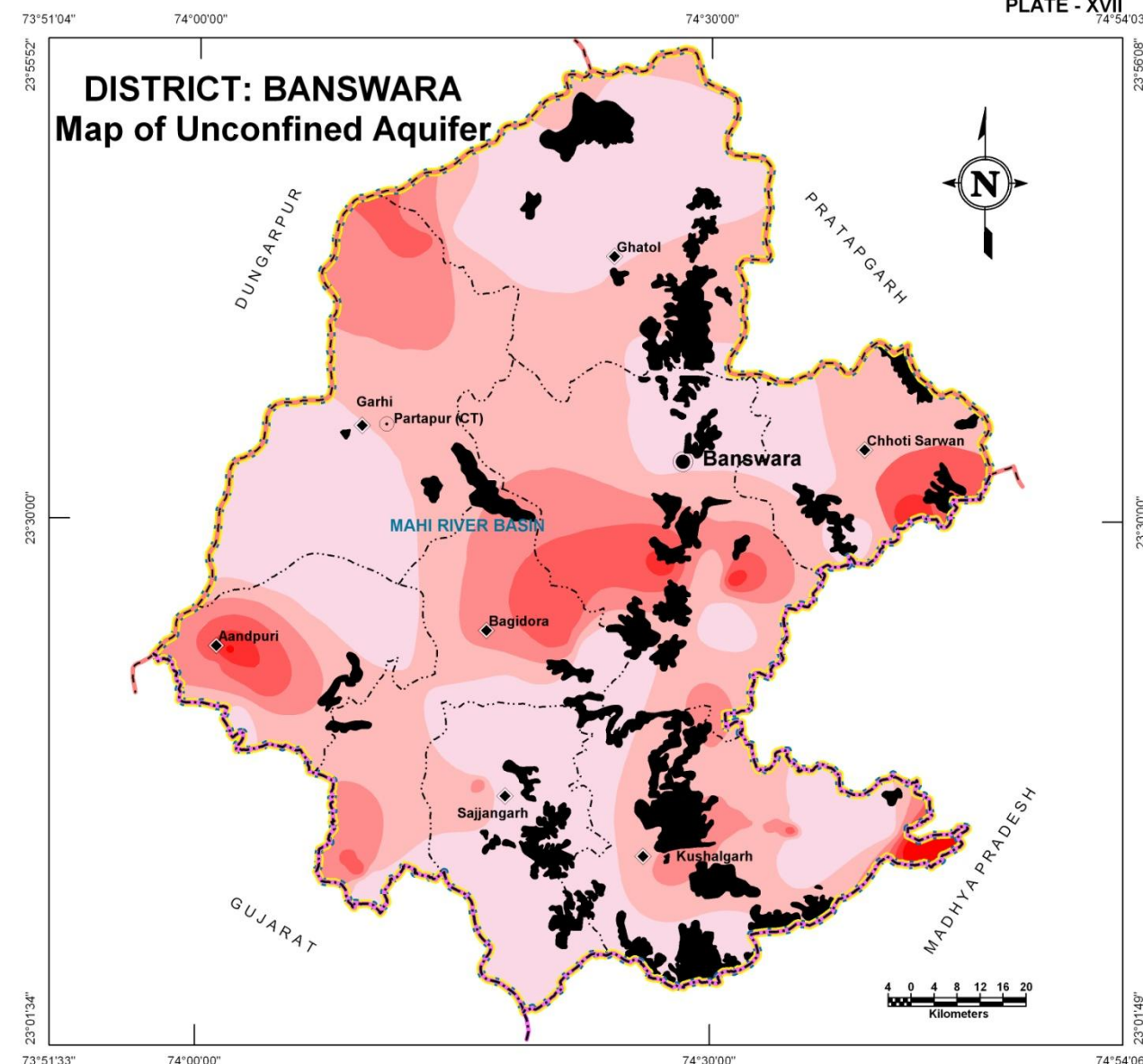
Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Hills

Depth to Bedrock(m bgl):

- < 20
- 20-40
- 40-60
- 60-80
- 80-100
- 100-120
- > 120

PLATE - XVII



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Hills

Aquifer Thickness(m):

- Weathered & Fractured Zone in Hardrock Area**
- <10
 - 10-20
 - 20-30
 - 30-40
 - 40-50
 - >50

Glossary of terms

S. No.	Technical Terms	Definition
1	AQUIFER	A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.
2	ARID CLIMATE	Climate characterized by high evaporation and low precipitation.
3	ARTIFICIAL RECHARGE	Addition of water to a ground water reservoir by man-made activity
4	CLIMATE	The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region.
5	CONFINED AQUIFER	A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.
6	CONTAMINATION	Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.
7	DRAWDOWN	The drawdown is the depth by which water level is lowered.
8	FRESH WATER	Water suitable for drinking purpose.
9	GROUND WATER	Water found below the land surface.
10	GROUND WATER BASIN	A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.
11	GROUND WATER RECHARGE	The natural infiltration of surface water into the ground.
12	HARD WATER	The water which does not produce sufficient foam with soap.
13	HYDRAULIC CONDUCTIVITY	A constant that serves as a measure of permeability of porous medium.
14	HYDROGEOLOGY	The science related with the ground water.
15	HUMID CLIMATE	The area having high moisture content.
16	ISOHYET	A line of equal amount of rainfall.
17	METEOROLOGY	Science of the atmosphere.
18	PERCOLATION	It is flow through a porous substance.
19	PERMEABILITY	The property or capacity of a soil or rock for transmitting water.
20	pH	Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).
21	PIEZOMETRIC HEAD	Elevation to which water will rise in a piezometers.
22	RECHARGE	It is a natural or artificial process by which water is added from outside to the aquifer.
23	SAFE YIELD	Amount of water which can be extracted from ground water without producing undesirable effect.
24	SALINITY	Concentration of dissolved salts.
25	SEMI-ARID	An area is considered semiarid having annual rainfall between 10-20 inches.
26	SEMI-CONFINED AQUIFER	Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.
27	SPECIFIC YIELD	Quantity of water which is released by a formation after its complete saturation.
28	TOTAL DISSOLVED SOLIDS	Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.

(Contd...)

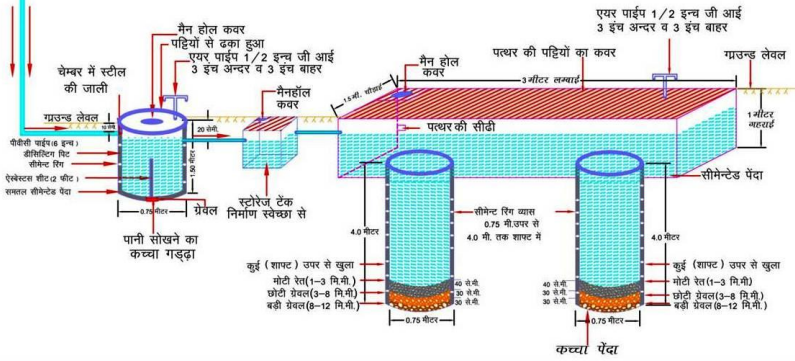
S. No.	Technical Terms	Definition
29	TRANSMISSIBILITY	It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability.
30	UNCONFINED AQUIFER	A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.
31	UNSATURATED ZONE	The zone below the land surface in which pore space contains both water and air.
32	WATER CONSERVATION	Optimal use and proper storage of water.
33	WATER RESOURCES	Availability of surface and ground water.
34	WATER RESOURCES MANAGEMENT	Planned development, distribution and use of water resources.
35	WATER TABLE	Water table is the upper surface of the zone of saturation at atmospheric pressure.
36	ZONE OF SATURATION	The ground in which all pores are completely filled with water.
37	ELECTRICAL CONDUCTIVITY	Flow of free ions in the water at 25C $\mu\text{m}/\text{cm}$.
38	CROSS SECTION	A Vertical Projection showing sub-surface formations encountered in a specific plane.
39	3-D PICTURE	A structure showing all three dimensions i.e. length, width and depth.
40	GWD	Ground Water Department
41	CGWB	Central Ground Water Board
42	CGWA	Central Ground Water Authority
43	SWRPD	State Water Resources Planning Department
44	EU-SPP	European Union State Partnership Programme
45	TOPOGRAPHY	Details of drainage lines and physical features of land surface on a map.
46	GEOLOGY	The science related with the Earth.
47	GEOMORPHOLOGY	The description and interpretation of land forms.
48	PRE MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)
49	POST-MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)
50	PIEZOMETER	A non-pumping small diameter bore hole used for monitoring of static water level.
51	GROUND WATER FLUCTUATION	Change in static water level below ground level.
52	WATER TABLE	The static water level found in unconfined aquifer.
53	DEPTH OF BED ROCK	Hard & compact rock encountered below land Surface.
54	G.W. MONITORING STATION	Dug wells selected on grid basis for monitoring of state water level.
55	EOLIAN DEPOSITS	Wind-blown sand deposits



भवन छत क्षेत्रफल 300 से 500 वर्गमीटर तक
निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

चित्र-4

- PVC पाईप 6" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.50 मी गहरा)
- रीचार्ज टैंक 1.5 मी चौड़ा x 3 मी लम्बा x 1 मी गहरा
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा) (संख्या 2)
- संरचना की अनुमानित लागत ₹ 24,000 अधिक
- वार्षिक पुनर्भरित जल लगभग 2,00,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 40,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



भूजल में घुले मुख्य तत्वों की अधिकता का मानव शरीर पर दुष्प्रभाव

बोरोन-स्नायु तन्त्र पर प्रभाव

फ्लोराइड - दंत क्षरण

क्लोराइड-सोडियम के साथ मिलकर उच्च रक्त चाप

सोडियम-हृदय, गुर्दा व रक्त परिसंचरण रोगों से ग्रसित लोगों को हानिकारक

कैल्शियम-जोड़ों में कड़ापन

नाइट्रेट-नवजात शिशुओं में ब्लू बेबी बीमारी (मेथेमोग्लोबिनिमिया)

आर्सेनिक-त्वचा रोग, कैंसर

सल्फेट-अधिकता में मैग्नेशियम के साथ मिलकर दस्तावर

लेड-बच्चों के शारीरिक व मानसिक विकास में बाधा वयस्कों में गुर्दे के रोग

आयरन-आयरन जीवाणु से आमाशय संबंधी रोग

फ्लोराइड-जोड़ों में अकड़न, हड्डियों में मुड़ाव



केंद्रीय भूमि जल बोर्ड,
पश्चिमी क्षेत्र, जयपुर
जल संसाधन मंत्रालय
भारत सरकार
e-mail: cgwbwr@sancharnet.in



भूजल अमूल्य है इसे प्रदूषित न करें।

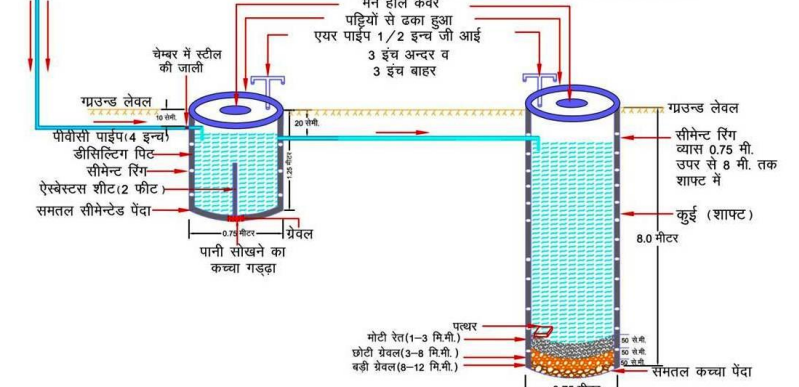


भवन छत क्षेत्रफल 100 से 200 वर्गमीटर तक

चित्र-2

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

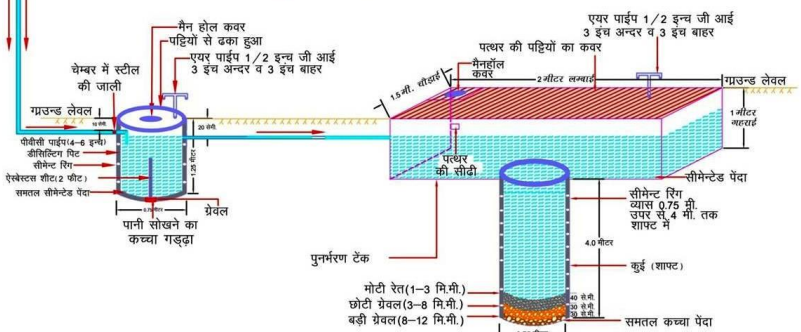
- PVC पाईप 4" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत ₹ 11,000-12,000
- वार्षिक पुनर्भरित जल लगभग 83,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 16,64,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



भवन छत क्षेत्रफल 200 से 300 वर्गमीटर तक
निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

चित्र-3

- PVC पाईप 4" - 6" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- रीचार्ज टैंक 1.5 मी चौड़ा x 2 मी लम्बा x 1 मी गहरा
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत ₹ 15,000-16,000
- वार्षिक पुनर्भरित जल लगभग 1,25,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 25,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम

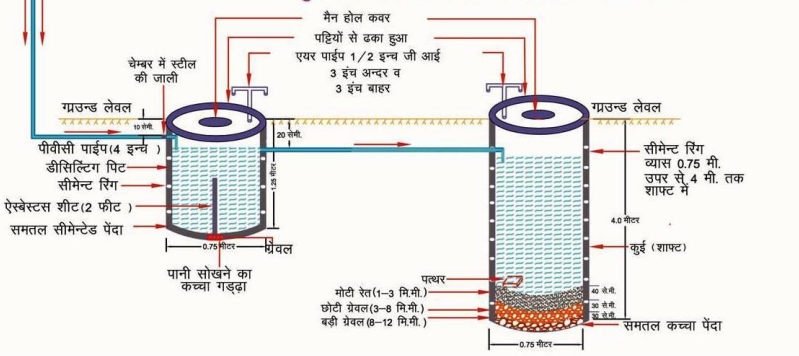


भवन छत क्षेत्रफल 100 वर्गमीटर तक

चित्र-1

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

- PVC पाईप 4" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत ₹ 7000-8000
- वार्षिक पुनर्भरित जल लगभग 40,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 8,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



Myths and Facts about Ground Water

S No	Myths	Facts
1	What is Ground Water • an underground lake • a net work of underground rivers • a bowl filled with water	Water which occurs below the land in geological formations/rocks is Ground water
2	Ground Water occurs everywhere beneath the Land Surface	Not really, it depends on the nature of rock formation
3	There is a relationship between ground water and surface water	Not all the places. Near streams/rivers there is relation
4	Groundwater is not renewable resource	It is renewable source and every year it is being recharged through rain/applied irrigation etc
5	Ground water is unlimited and deeper you drill more discharge	It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper
6	Ground Water moves rapidly	The movement of ground water is very slow
7	Ground water pumped from wells is thousands of years old	Generally the ground water being tapped through wells is a few years old
8	If water taste good—it is safe to drink	It may have other chemicals e.g. fluoride, nitrates etc which are harmful
9	Water from free flowing tube wells is very pure	This water can also be contaminated so test before use
10	If I recharge my TW/DW/HP it will not benefit me	It will also benefit you and also adjoining wells
11	There is no static ground water resources in Rajasthan	Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge
12	I cannot meet annual cooking and drinking water requirement by rain water harvesting	The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement
13	You can increase ground water recharge	This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures
14	You cannot use abandoned TW/HP/DW for ground water recharge	These should be used as recharge structures as harvested rain water is directly put into GW reservoir
15	Putting waste near HP/TW will not cause any problem	Such actions will pollute wells and water



ROLTA

Rolta India Limited

Central & Registered Office

Rolta Tower A,

Rolta Technology Park,

MIDC, Andheri (East),

Mumbai - 400 093

Tel : +91 (22) 2926 6666, 3087 6543

Fax : +91 (22) 2836 5992

Email : indsales@rolta.com

www.rolta.com